

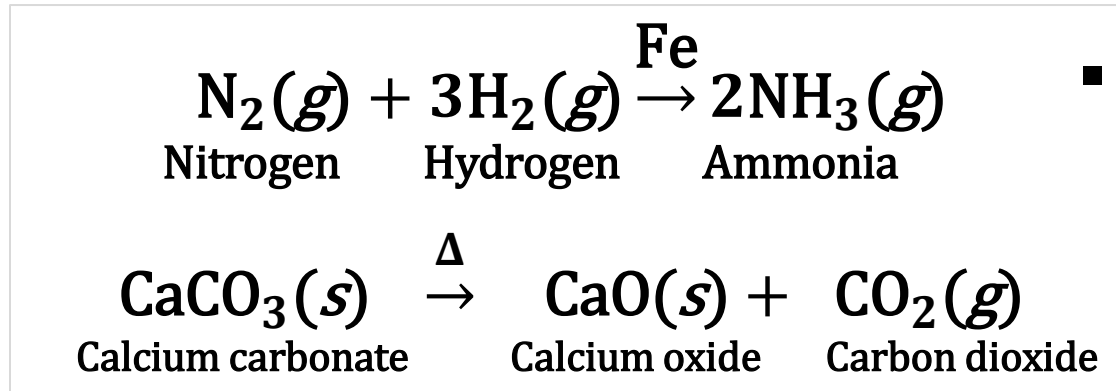
Chapter 14

Chemical Kinetics

- ▶ Factors That Affect Reaction Rates
- ▶ Reaction Rates
- ▶ Concentration and Rate Laws
- ▶ The Change of Concentration with Time
- ▶ Temperature and Rate
- ▶ Reaction Mechanisms
- ▶ Catalysis

Chemical Kinetics

Introduction



- Reading a reaction equation is important.
 - ✓ State, Condition, Stoichiometry

Q. Do all chemical reactions progress at the **same speed**?

Q. Does "spontaneous" mean fast?

- The speed at which reactions take place is called the **reaction rate**.
- The study of reaction rate is called **chemical kinetics**.
- A step-by-step view of the change of reactants to products is called a **mechanism**.

Chemical Kinetics

Factors That Affect Reaction Rates

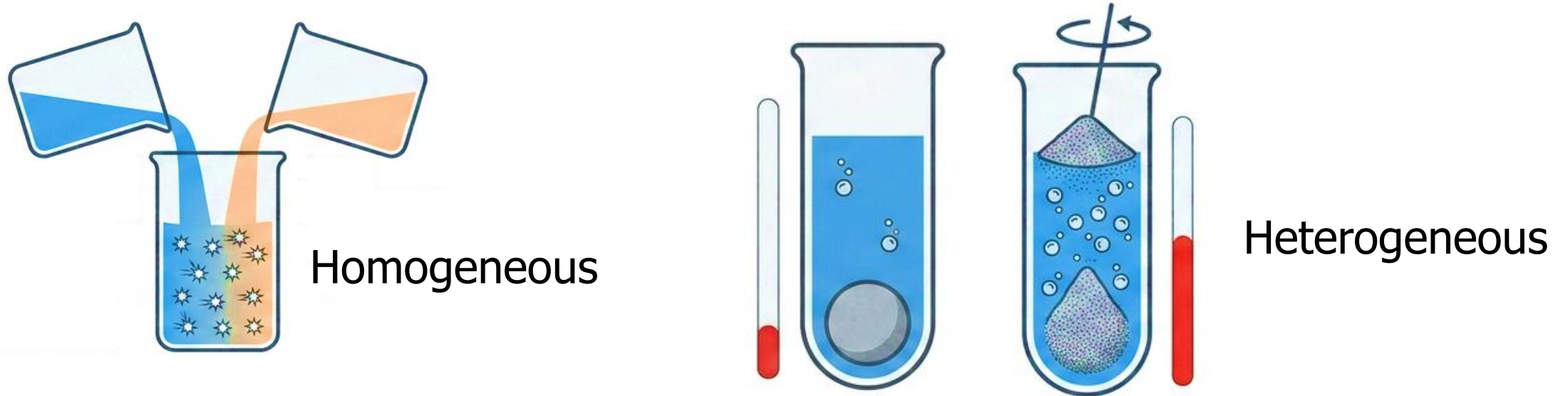
Q. What factors affect the rate of a chemical reaction?

1. Physical state of the reactants
2. Reactant concentrations
3. Reaction temperature
4. Presence of a catalyst

Factors That Affect Reaction Rates

1. Physical State of the Reactants

- The more readily the reactants collide, the more rapidly they react.
 - ✓ Homogeneous reactions (all gases or liquids) are often faster.
 - ✓ Heterogeneous reactions that involve solids are slower.



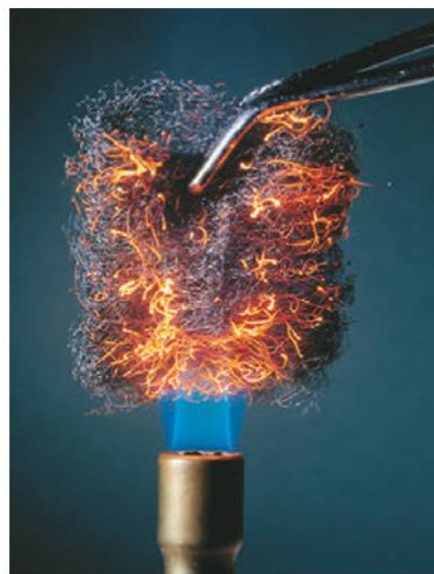
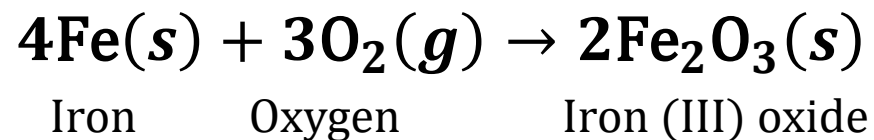
Q. How can the reactivity of solid substances be increased?

Factors That Affect Reaction Rates

2. Reactant Concentrations

- Increasing reactant concentration generally increases reaction rate.
- Since there are more molecules, more collisions occur.

Q. If a heated steel nail were placed in pure O_2 , would you expect it to burn as readily as the steel wool does?



Steel wool heated in air (about 20% O_2) glows red-hot but oxidizes to Fe_2O_3 slowly.



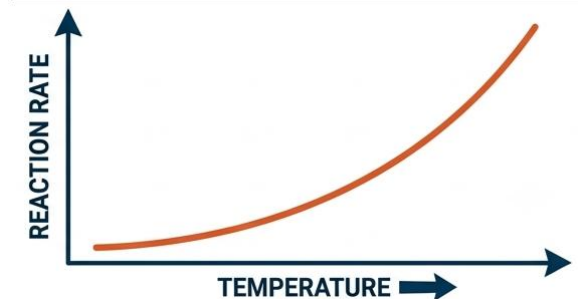
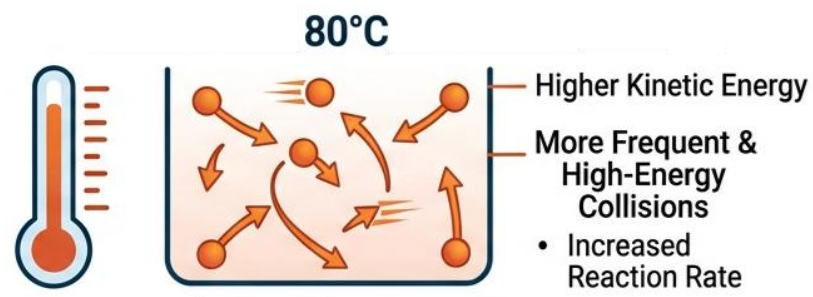
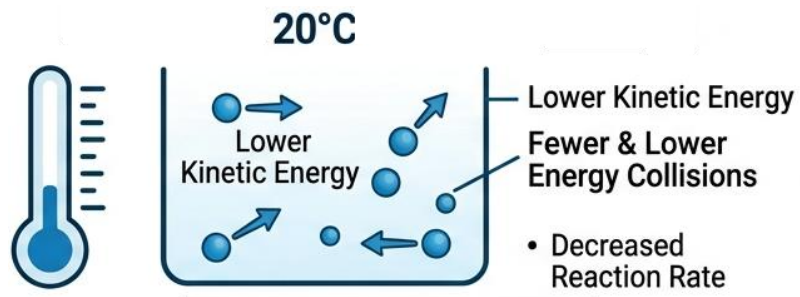
Red-hot steel wool in 100% O_2 burns vigorously, forming Fe_2O_3 quickly.

Effect of concentration on reaction rate.

Factors That Affect Reaction Rates

3. Temperature

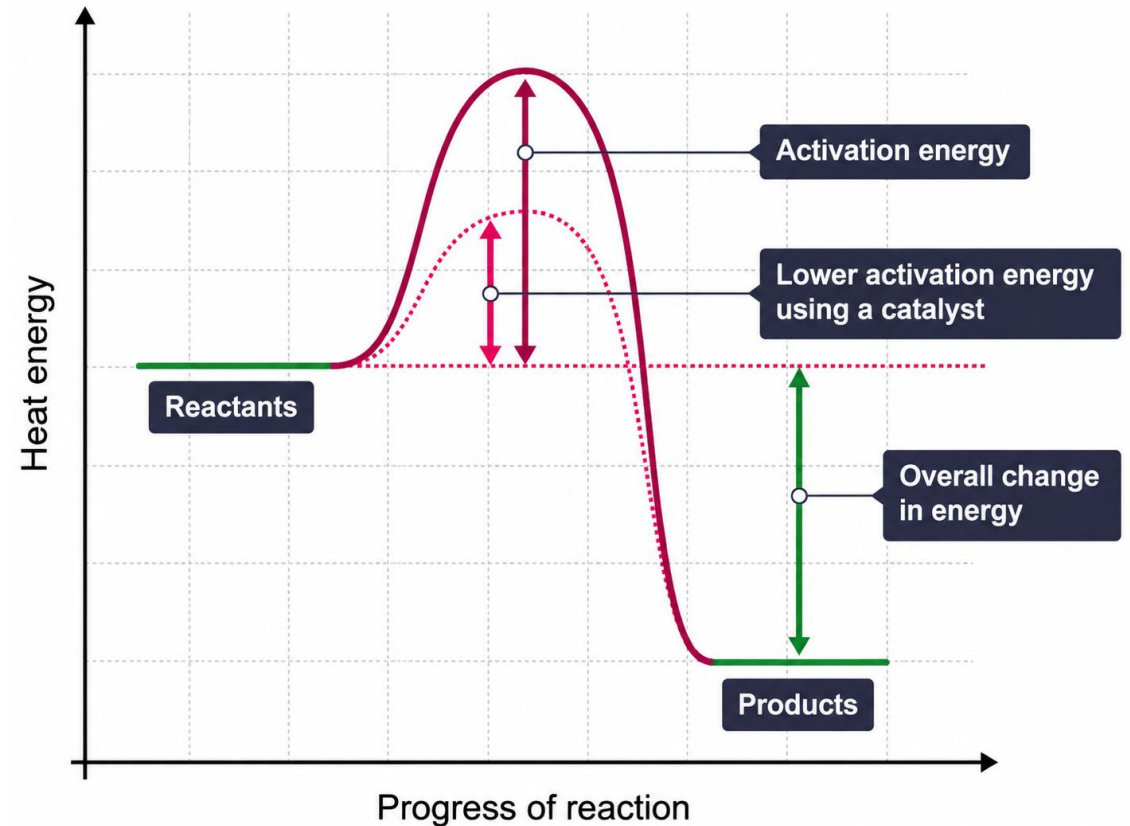
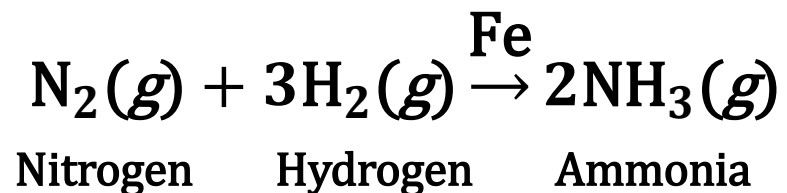
- Reaction rate generally increases with increased temperature.
- The kinetic energy of molecules is related to temperature.
- At higher temperatures, molecules move more quickly, increasing the number of collisions and the energy the molecules possess during the collisions.



Factors That Affect Reaction Rates

4. Presence of a Catalyst

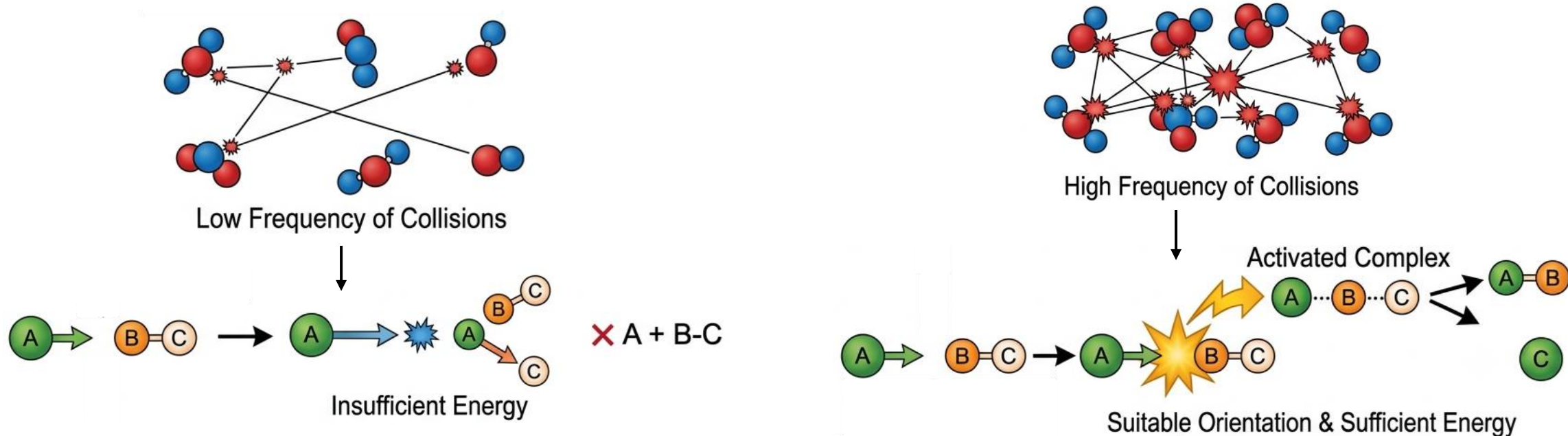
- Catalysts are agents that increase reaction rates without themselves being used up.
- Catalysts affect the kinds of collisions, changing the mechanism (individual reactions that are part of the pathway from reactants to products).
- Catalysts are critical in many biological reactions.



Factors That Affect Reaction Rates

Chemical reactions at a molecular level

- Reaction rates depend on the **frequency of collisions** between molecules.

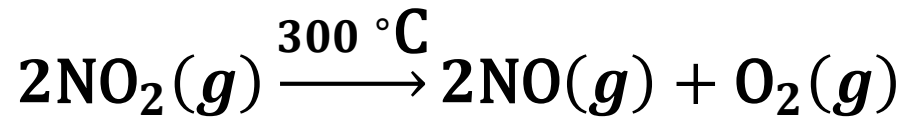


- A greater frequency of collisions leads to a higher reaction rate.
 - ✓ Collisions must occur with sufficient energy to break bonds.
 - ✓ Collisions must lead to **suitable orientation** for new bonds to form.

Reaction Rate

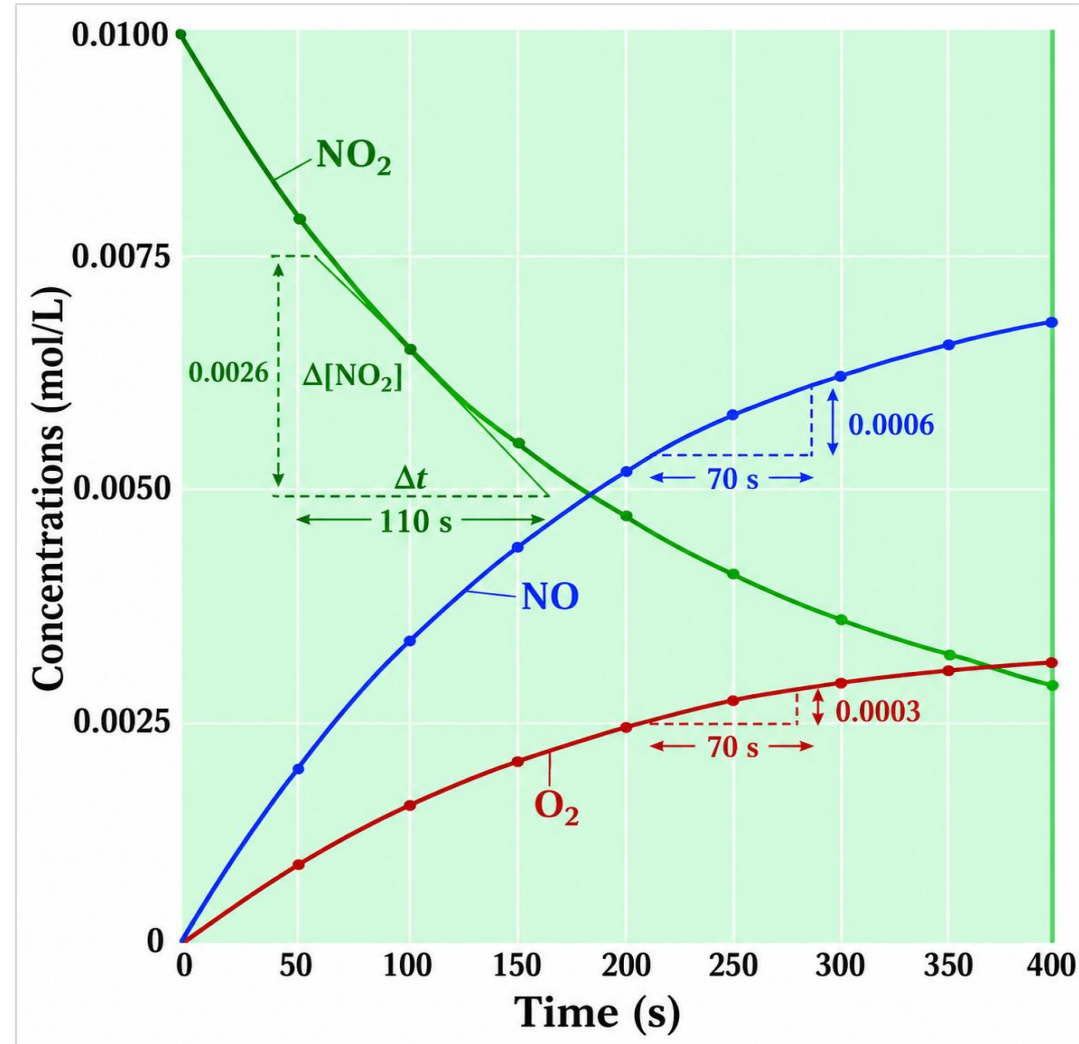
Q. How do we define the rate of a chemical reaction?

Example



Nitrogen dioxide Nitric oxide Oxygen

- Starting with a flask of nitrogen dioxide at 300°C, the concentrations of nitrogen dioxide, nitric oxide, and oxygen are plotted versus time.



Reaction Rate

Q. How do we define the rate of a chemical reaction?

- The rate of a reaction, the speed at which it occurs, is the change in the concentration of reactants or products **per unit of time**.

$$\text{Rate} = \frac{\Delta[\]}{\Delta t}$$

- ✓ Δ : “change in”
- ✓ $[\]$: molar concentration (M)
- ✓ t : time (s)



$$\text{Rate} = \frac{\Delta[P]}{\Delta t}$$

$$\text{Rate} = - \frac{\Delta[R]}{\Delta t}$$

Note: Rates are always expressed as positive quantities.

Q. What is the unit of rate?

Reaction Rate

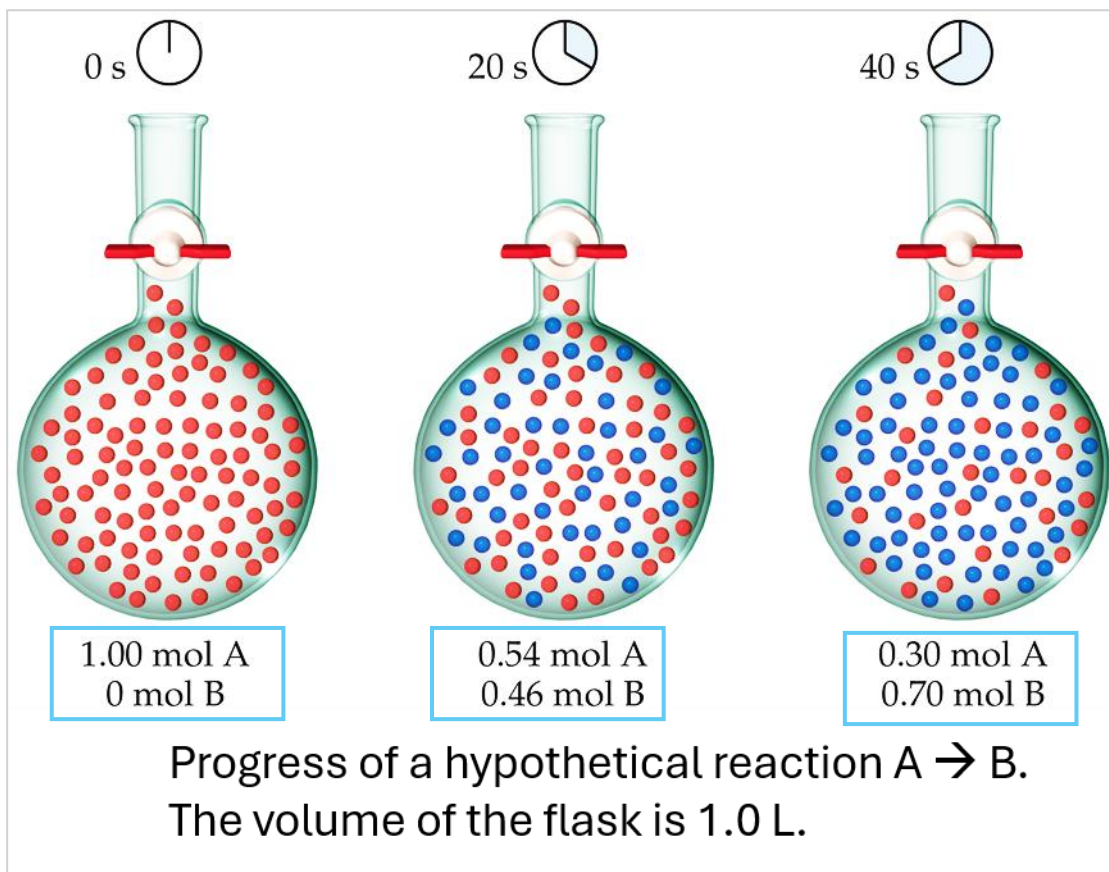
Q. How do we define the rate of a chemical reaction?

Type of rate	Definition	How it is determined
1. Average rate	Change in concentration over a time interval	$\frac{\Delta[\text{concentration}]}{\Delta t}$
2. Instantaneous rate	Rate of concentration change at a specific instant	Slope of the tangent line to the concentration–time curve
3. Initial rate	Instantaneous rate at $t = 0$	Slope of the tangent to the curve at the beginning of the reaction (concentration–time curve)

Let's see Examples!

Reaction Rate: Following Reaction Rates

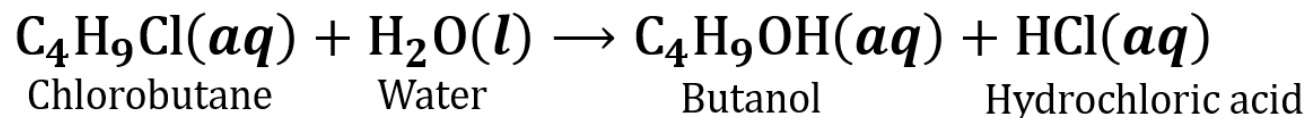
Example-1:- Average rate



Q. Determine the average rates of **appearance** of B and **disappearance** of A over the 20 s interval from the beginning of the reaction. (Ans.: $2.3 \times 10^{-2} \text{ M/s}$ (each)).

Reaction Rate: Following Reaction Rates

Example-2:- Average rate



Rate data for the reaction of $\text{C}_4\text{H}_9\text{Cl}$ with water

Time, t (s)	$[\text{C}_4\text{H}_9\text{Cl}]$ (M)	Average rate (M/s)
0.0	0.1000	-
50.0	0.0905	1.9×10^{-4}
100.0	0.0820	1.7×10^{-4}
150.0	0.0741	1.6×10^{-4}
200.0	0.0671	1.4×10^{-4}
300.0	0.0549	1.22×10^{-4}
400.0	0.0448	1.01×10^{-4}
500.0	0.0368	0.80×10^{-4}
800.0	0.0200	0.560×10^{-4}
10,000	0	

- Reactions typically slow down over time!
Why?

Q. Confirm all the average rates in the table using the following equation.

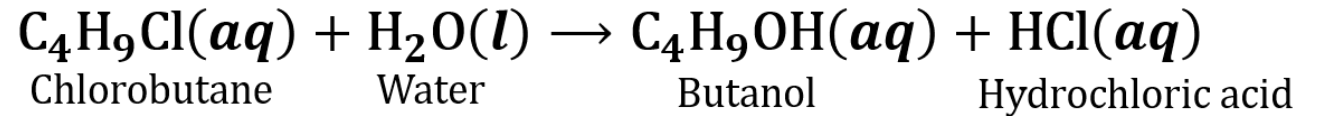
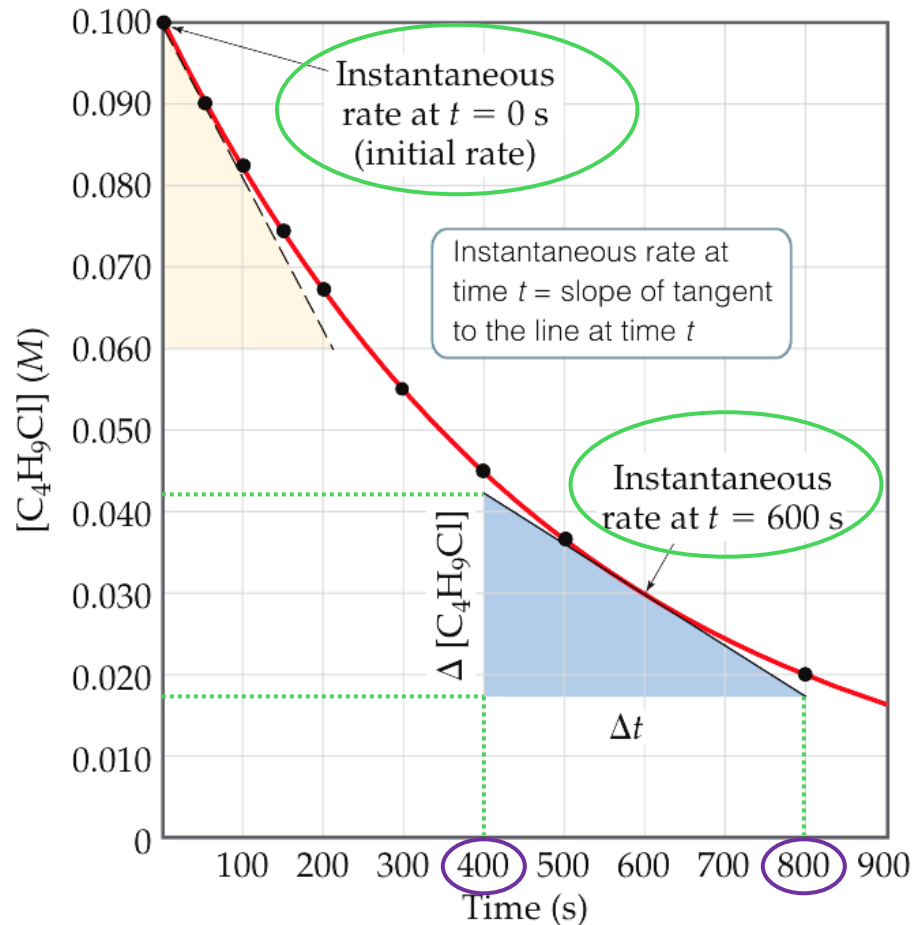
$$\text{Average rate of disappearance} = - \frac{[\text{C}_4\text{H}_9\text{Cl}]_{t_2} - [\text{C}_4\text{H}_9\text{Cl}]_{t_1}}{t_2 - t_1}$$

For example, from 50.0 to 100.0 s:

$$- \frac{0.0820 - 0.0905}{100.0 - 50.0} = 1.70 \times 10^{-4} \text{ M/s}$$

Reaction Rate: Plotting Rate Data

Example-3:- Instantaneous rate and Initial rate



Q. Determine the instantaneous rate at 600 s

$$\text{Instantaneous rate} = -\frac{\Delta[C_4H_9Cl]}{\Delta t} = -\frac{(0.017 - 0.042) M}{(800 - 400) s}$$

$$= 6.3 \times 10^{-5} M/s$$

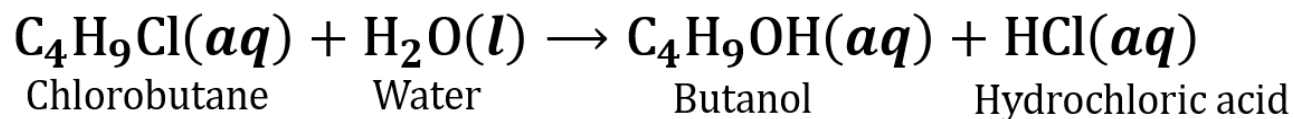
Q. Determine the rate of disappearance of chlorobutane at $t = 0$ s (initial rate) from the graph.

Concentration of C_4H_9Cl as a function of time.

Reaction Rate: Relative Rates and Stoichiometry

Q. How does the stoichiometry of a reaction affect the expression for its reaction rate?

Consider:



- **Stoichiometry:** 1 mol of $\text{C}_4\text{H}_9\text{OH}$ is produced for each mole of $\text{C}_4\text{H}_9\text{Cl}$ consumed (the ratio of $\text{C}_4\text{H}_9\text{Cl}$ to $\text{C}_4\text{H}_9\text{OH}$ is 1:1).

$$\text{Rate} = -\frac{\Delta[\text{C}_4\text{H}_9\text{Cl}]}{\Delta t} = \frac{\Delta[\text{C}_4\text{H}_9\text{OH}]}{\Delta t}$$

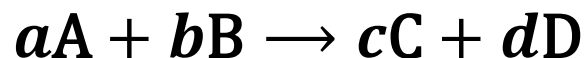
Rate of disappearance of $\text{C}_4\text{H}_9\text{Cl}$ = Rate of appearance of $\text{C}_4\text{H}_9\text{OH}$

Q. What if the equation (stoichiometry) is *not* 1:1?

Reaction Rate: Relative Rates and Stoichiometry

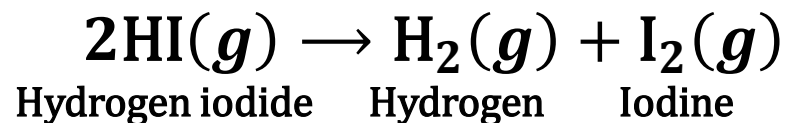
Q. What if the equation (stoichiometry) is *not* 1:1?

- For any general reaction:



$$\text{Rate} = -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}$$

- Example:**



$$\text{Rate} = -\frac{1}{2} \frac{\Delta[\text{HI}]}{\Delta t} = \frac{\Delta[\text{H}_2]}{\Delta t} = \frac{\Delta[\text{I}_2]}{\Delta t}$$

- At a given instant, a reaction has one defined reaction rate.
- Different species may show different concentration-change rates.
- Stoichiometric coefficients normalize these individual rates to give the same reaction rate.

Reaction Rate: Relative Rates and Stoichiometry

Consider the reaction:

$2\text{O}_3(g)$	$\rightarrow$	$3\text{O}_2(g)$
Ozone		Oxygen

Q. How is the rate at which ozone disappears related to the rate at which oxygen appears in the reaction?

$$\text{Rate} = -\frac{1}{2} \frac{\Delta[\text{O}_3]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{O}_2]}{\Delta t}$$

Q. If the rate at which O_2 appears is $6.0 \times 10^{-5} \text{ M/s}$ at a particular instant, at what rate is O_3 disappearing at this same time?

$$-\frac{\Delta[\text{O}_3]}{\Delta t} = \frac{2}{3} \frac{\Delta[\text{O}_2]}{\Delta t} = \frac{2}{3} (6.0 \times 10^{-5} \text{ M/s}) = 4.0 \times 10^{-5} \text{ M/s}$$

Concentration and Rate Laws

Q: What factors affect the rate of a chemical reaction?

1. Physical state of the reactants
- 2. Reactant concentrations**
3. Reaction temperature
4. Presence of a catalyst

Q. How do we determine what effect the concentration of each reactant has on the rate of the reaction?

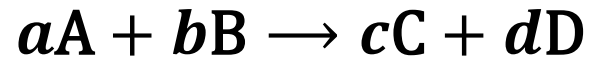
- **Rate Law:** An **equation** that shows how the rate depends on reactant concentrations.

$$\text{Rate} = k[A]^m[B]^n$$

Concentration and Rate Laws

Q. How do we write the Rate Law equation?

- For a general reaction:



$$\text{Rate} = k[A]^m[B]^n$$

- k**: rate constant
- m** and **n** : Reaction Orders (integer or fraction) with respect to each reactant

$$\text{Rate} = k[\text{Reactant 1}]^m[\text{Reactant 2}]^n...$$

- Reaction orders and rate constants must be determined **by experiment**
- Reaction orders: indicate how the rate is affected by each reactant concentration.
- Overall reaction order: the sum of the orders with respect to each reactant ($m + n + \dots$).
- Reaction orders $\neq$ the coefficients in the balanced equation
- Determining the rate law means determining the reaction order with respect to each reactant.

Concentration and Rate Laws: The Exponents in the Rate Law

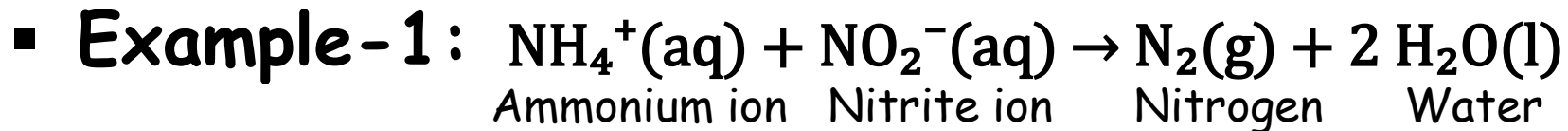


TABLE 14.2 Rate Data for the Reaction of Ammonium and Nitrite Ions in Water at 25 °C

Experiment	Initial $[\text{NH}_4^+]$ (M)	Initial $[\text{NO}_2^-]$ (M)	Initial rate (M/s)
1	0.0100	0.200	5.4×10^{-7}
2	0.0200	0.200	10.8×10^{-7}
3	0.0400	0.200	21.5×10^{-7}
4	0.200	0.0202	10.8×10^{-7}
5	0.200	0.0404	21.6×10^{-7}
6	0.200	0.0808	43.3×10^{-7}

Experiment 1 to 3

$[\text{NO}_2^-]$ is constant. How does $[\text{NH}_4^+]$ affect the rate?

$$\text{Rate} \propto [\text{NH}_4^+]$$

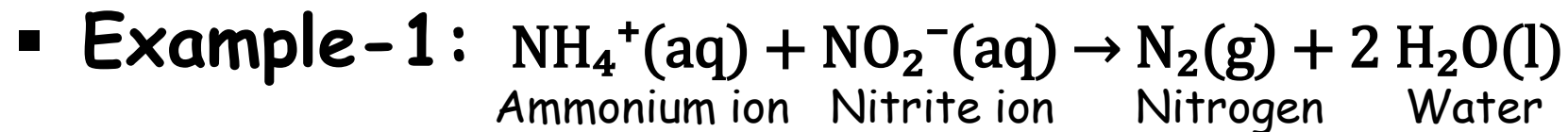
Experiment 4 to 6

$[\text{NH}_4^+]$ is constant. How does $[\text{NO}_2^-]$ affect the rate?

$$\text{Rate} \propto [\text{NO}_2^-]$$

$$\text{Rate} = k [\text{NH}_4^+]^1 [\text{NO}_2^-]^1 = k [\text{NH}_4^+] [\text{NO}_2^-]$$

Concentration and Rate Laws: The Exponents in the Rate Law



Rate = $k[\text{NH}_4^+][\text{NO}_2^-]$
 ✓ First order with respect to each reactant.
 ✓ Second order overall ($1 + 1 = 2$).

General reaction	Rate law	Order	If [A] is doubled	If [A] is tripled
$A \rightarrow B$	$\text{Rate} = k[A]^1$	First order	$2 \times \text{rate}$	$3 \times \text{rate}$
$A \rightarrow B$	$\text{Rate} = k[A]^2$	Second order	$4 \times \text{rate}$	$9 \times \text{rate}$
$A \rightarrow B$	$\text{Rate} = k[A]^0$	Zero order	No change	No change

Q. If [A] is doubled and the initial rate becomes four times larger, what is the order with respect to A?

Concentration and Rate Laws: The Exponents in the Rate Law

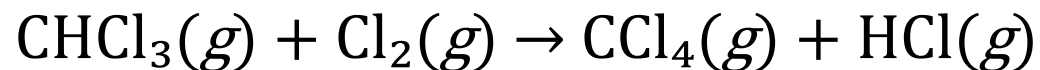
- **Additional examples: Experimentally determined rate laws**



$$\text{Rate} = k [\text{N}_2\text{O}_5]$$



$$\text{Rate} = k [\text{H}_2][\text{I}_2]$$



$$\text{Rate} = k [\text{CHCl}_3][\text{Cl}_2]^{1/2}$$

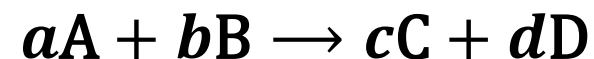
Q. Identify the order with respect to each reactant and the overall reaction order for each reaction.

Magnitudes and Units of Rate Constants

Determination of k (the rate constant)

Requirements to Determine k :

- ✓ Rate law of the reaction is known
 - ✓ Reaction rate is known
 - ✓ Reactant concentrations are known
- k is often used to determine the relative rate of a reaction.
 - **The units of the rate constant depend on the overall reaction order.**



$$\text{Rate} = k[A]^m[B]^n$$

$$k = \frac{\text{Rate}}{[A]^m[B]^n}$$

Magnitudes and Units of Rate Constants

Determination of k : Example (Table 14.2)

- Using the values for experiment 1 for: $\text{NH}_4^+(\text{aq}) + \text{NO}_2^-(\text{aq}) \rightarrow \text{N}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$
Ammonium ion Nitrite ion Nitrogen Water

Experiment	Initial $[\text{NH}_4^+]$ (M)	Initial $[\text{NO}_2^-]$ (M)	Initial rate (M/s)
1	0.0100	0.200	5.4×10^{-7}

$$\text{Rate} = k[\text{NH}_4^+][\text{NO}_2^-]$$

$$k = \text{Rate} / [\text{NH}_4^+][\text{NO}_2^-]$$

$$k = [(5.4 \times 10^{-7} \text{ M/s}) / [(0.0100 \text{ M})(0.200 \text{ M})]$$

$$k = 2.7 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1} \quad (\text{M} = \text{molarity}; \text{s} = \text{seconds})$$

- For a second-order reaction, k has units $\text{M}^{-1} \text{s}^{-1}$.

Magnitudes and Units of Rate Constants

Q. Determine the unit of k for the following reactions.



$$\text{Rate} = k [\text{N}_2\text{O}_5]$$



$$\text{Rate} = k [\text{H}_2][\text{I}_2]$$

Q. Fill the missing information in the following table.

Reaction order	General rate equation	Unit of k	Example
First order			
Second order			
Zero order			

Using Initial Rates to Determine Rate Laws

Q. The initial rate of a reaction $A+B \rightarrow C$ was measured for three different starting concentrations of A and B, and the results are as follows:

Experiment Number	[A](M)	[B](M)	Initial Rate (M/s)
1	0.100	0.100	4.0×10^{-5}
2	0.100	0.200	4.0×10^{-5}
3	0.200	0.100	16.0×10^{-5}

Using these data, determine:

- (a) The rate law (i.e. determine reaction orders for each reactant).
- (b) The rate constant.
- (c) The rate when $[A] = 0.050 \text{ M}$ and $[B] = 0.100 \text{ M}$.

Text book: pp. 671 - 672

Integrated Rate Law: The Change of Concentration with Time

Q. How to express the relationship between the **concentrations** of reactants (or products) and **time**?



$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^n$$

- **Differential rate law:** Expresses how the rate depends on concentration.
- **Integrated rate law:** Expresses how the concentrations depend on time.

Q. Explain the key difference between these two rate-law forms.

Integrated Rate Law: The Change of Concentration with Time

Integrated Rate Law for **First-Order** Reactions: $A \rightarrow P$

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]$$

Differential rate law

- Integrate both sides to obtain the integrated rate law

$$\text{Rate} = -\frac{d[A]}{[A]} = kdt$$

Differential form

$$\ln[A]_t = -kt + \ln[A]_0$$

Integrated rate law

$[A]_t$: the concentration of A at time t

$[A]_0$: the initial concentration

$$y = mx + b$$

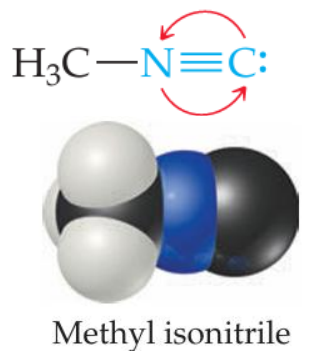
For a first-order:

- ✓ a plot of $\ln[A]_t$ versus time is linear
- ✓ slope = $-k$
- ✓ y-intercept = $\ln[A]_0$.

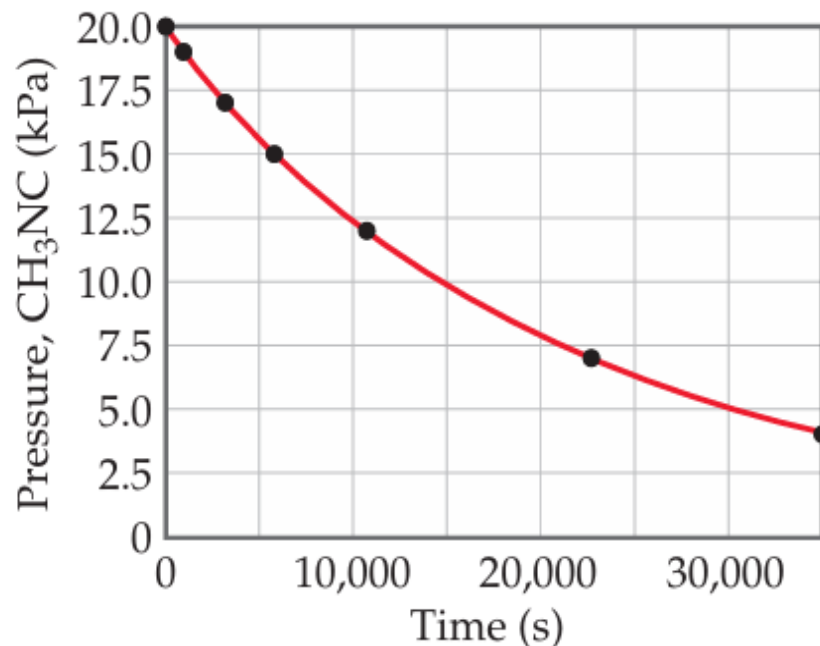
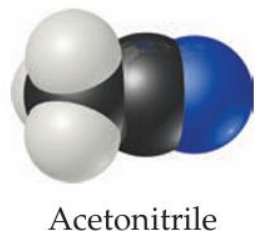
Q. Show how this integrated rate law is derived from the differential rate law.

Integrated Rate Law: The Change of Concentration with Time

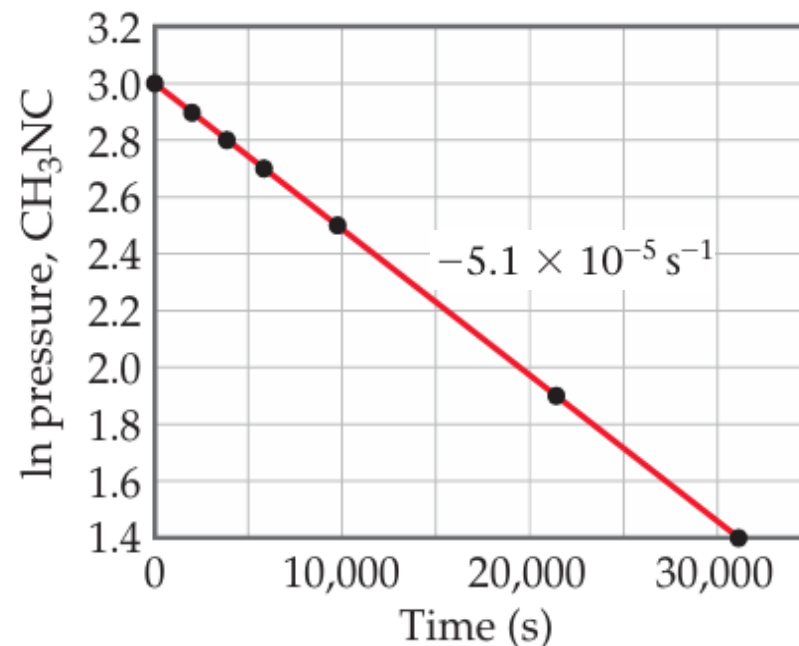
Example: **First-Order** Integrated Rate Law



199 °C



(a)



(b)

Kinetic data for conversion of methyl isonitrile into acetonitrile (first order)

$$\ln[\text{CH}_3\text{NC}]_t = -kt + \ln[\text{CH}_3\text{NC}]_0$$

Integrated Rate Law: The Change of Concentration with Time

Integrated Rate Law for **Second-Order** Reactions: $A \rightarrow P$

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^2$$

$$\text{Rate} = \frac{d[A]}{[A]^2} = -kdt$$

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

$$y = mx + b$$

Differential rate law

- Integrate both sides to obtain the integrated rate law

Differential form

Integrated rate law

$[A]_t$: the concentration of A at time t

$[A]_0$: the initial concentration

For a second-order:

- ✓ a plot of $1/[A]_t$ versus time is linear
- ✓ slope = k
- ✓ y-intercept = $1/[A]_0$.

Q. Show how this integrated rate law is derived from the differential rate law.

Integrated Rate Law: The Change of Concentration with Time

- **Remark:** One way to distinguish between first- and second-order rate laws is to graph both $\ln[A]_t$ and $1/[A]_t$ against t .

Q. The following data were obtained for the gas-phase decomposition of nitrogen dioxide at 300 °C. Is the reaction first or second order in NO_2 ?



Time (s)	$[\text{NO}_2]$ (M)
0	0.01000
50.0	0.00787
100.0	0.00649
200.0	0.00481
300.0	0.00380

Hint: Draw $\ln[A]_t$ and $1/[A]_t$ against t and check the slope. (pp. 677)

Integrated Rate Law: The Change of Concentration with Time

Integrated Rate Law for **Zero-Order** Reactions: $A \rightarrow P$

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^0 = k \quad \text{Differential rate law}$$

- Integrate both sides to obtain the integrated rate law

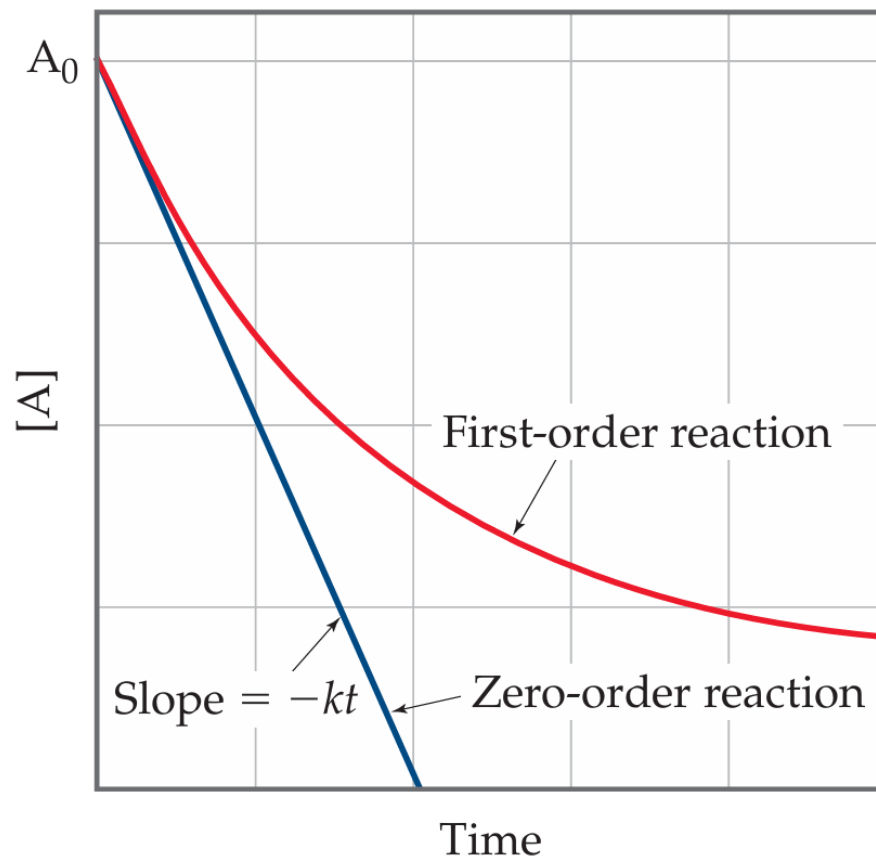
$$[A]_t = -kt + [A]_0 \quad \text{Integrated rate law}$$

$[A]_t$: the concentration of A at time t

$[A]_0$: the initial concentration

For a zero-order:

- ✓ Plot of $[A]_t$ versus time is linear
- ✓ Slope = $-k$
- ✓ y-intercept = $[A]_0$.



▲ **Figure 14.9** Comparison of first-order and zero-order reactions for the disappearance of reactant A with time.

Integrated Rate Law: The Change of Concentration with Time

■ Integrated rate laws: Summary

Reaction order	Integrated rate law
First-Order Reactions	$\ln[A]_t = -kt + \ln[A]_o$
Second-Order Reactions	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_o}$
Zero-Order Reactions	$[A]_t = -kt + [A]_o$

■ These equations can be used to determine

1. The concentration of a reactant remaining **at any time** after the reaction has started.
2. The time interval required for a given fraction of a sample to react.
3. The time interval required for a reactant concentration to fall to a certain level.

Integrated Rate Law: The Change of Concentration with Time

Half-Life ($t_{1/2}$): the time required for the concentration of a reactant to reach half its initial value.



$$[A]_{t_{1/2}} = \frac{1}{2} [A]_0$$

Reaction order	Integrated rate law	Half-life, $t_{1/2}$	Dependence of $t_{1/2}$ on $[A]_0$
Zero order	$[A]_t = -kt + [A]_0$	$t_{1/2} = \frac{[A]_0}{2k}$	Depends on $[A]_0$
First order	$\ln[A]_t = -kt + \ln[A]_0$	$t_{1/2} = \frac{0.693}{k}$	Independent of $[A]_0$
Second order	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$	$t_{1/2} = \frac{1}{k[A]_0}$	Depends inversely on $[A]_0$

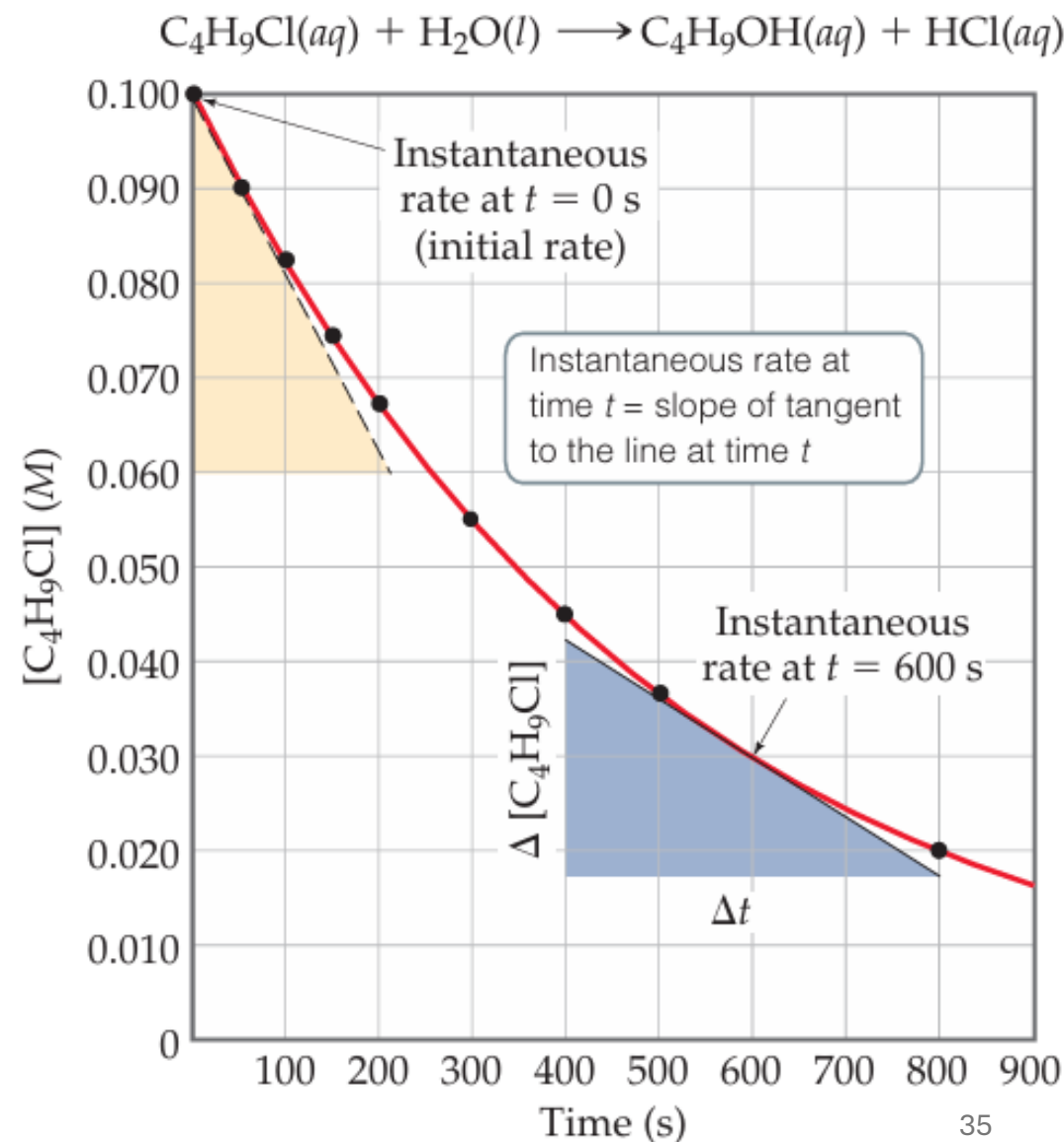
- Q.** How does $t_{1/2}$ change with the initial concentration of a reactant for each order?
- Q.** Show the derivation of half life equation from integrated equation for each order.

Integrated Rate Law: The Change of Concentration with Time

Q. The reaction of C_4H_9Cl with water is a **first-order** reaction.

(a) Estimate the half-life for this reaction (from the graph).

(b) Use the half-life equation to calculate the rate constant.



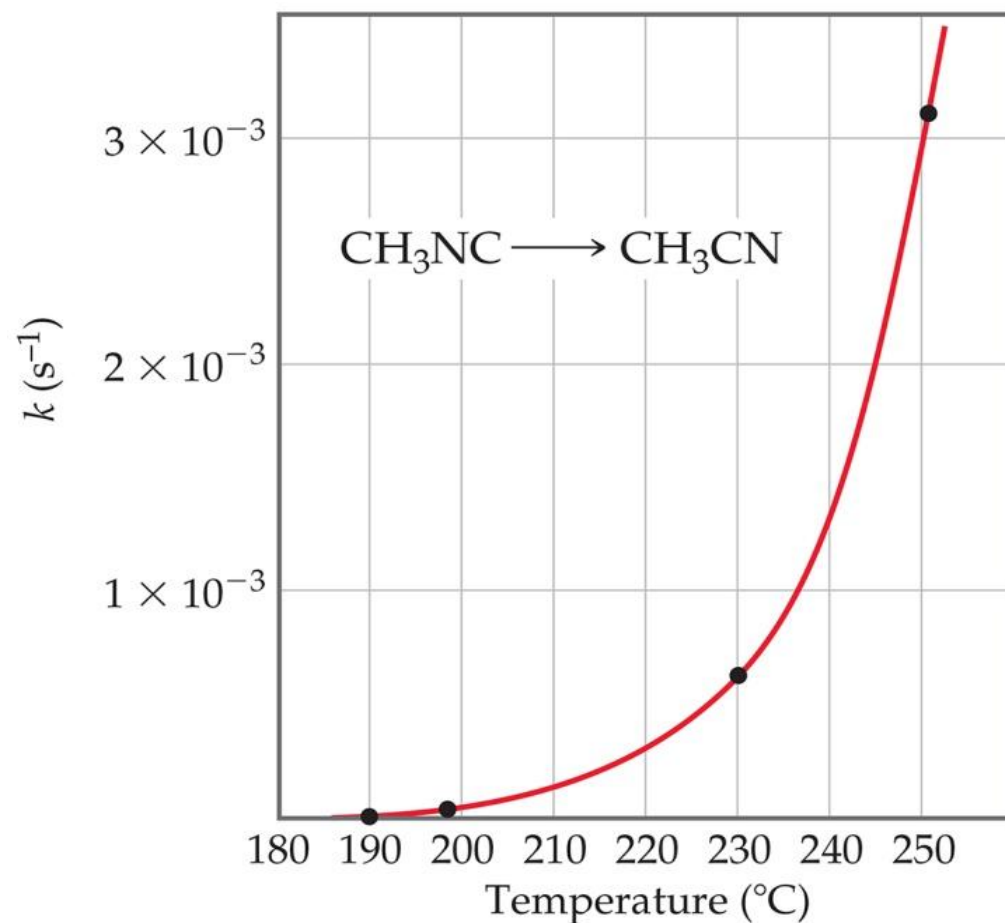
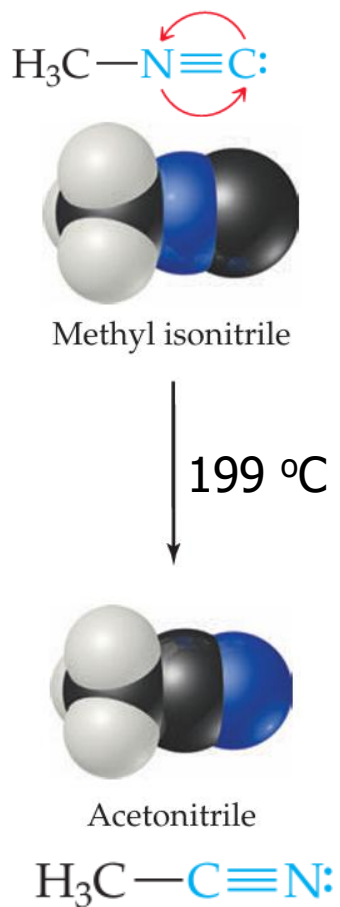
Temperature and Rate

Q. How does temperature affect the rate of a chemical reaction?

- As temperature increases, the rate increases.
- The rate constant (k) is temperature-dependent.
- k increases as temperature increases.
- k doubles (approximately) with every 10 °C rise.

Temperature and Rate

Example

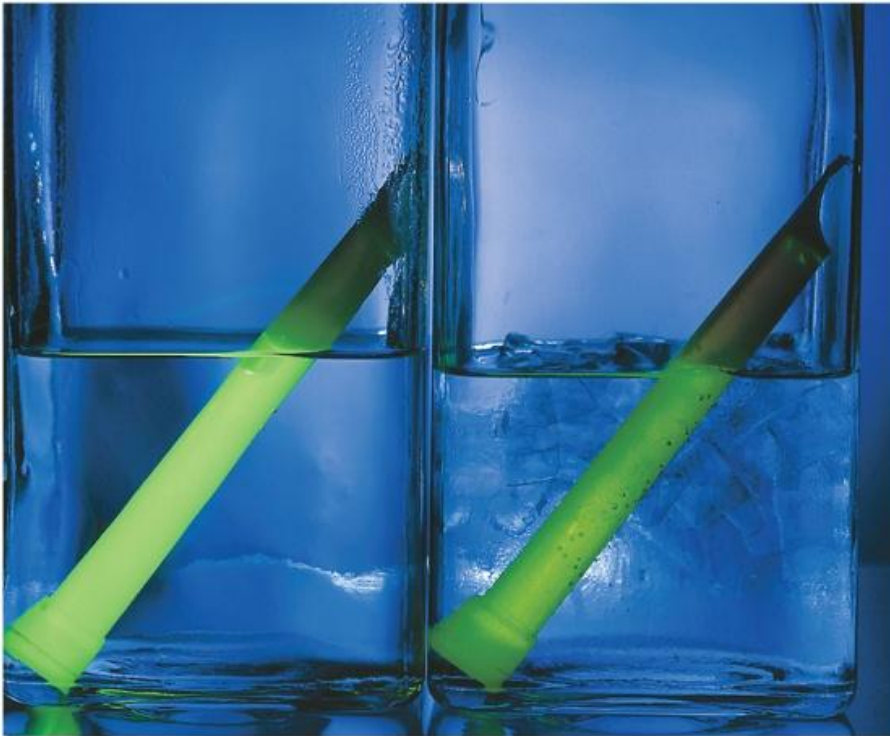


Q. Why is the increase non-linear?

Temperature dependence of the rate constant for methyl isonitrile conversion to acetonitrile

Temperature and Rate

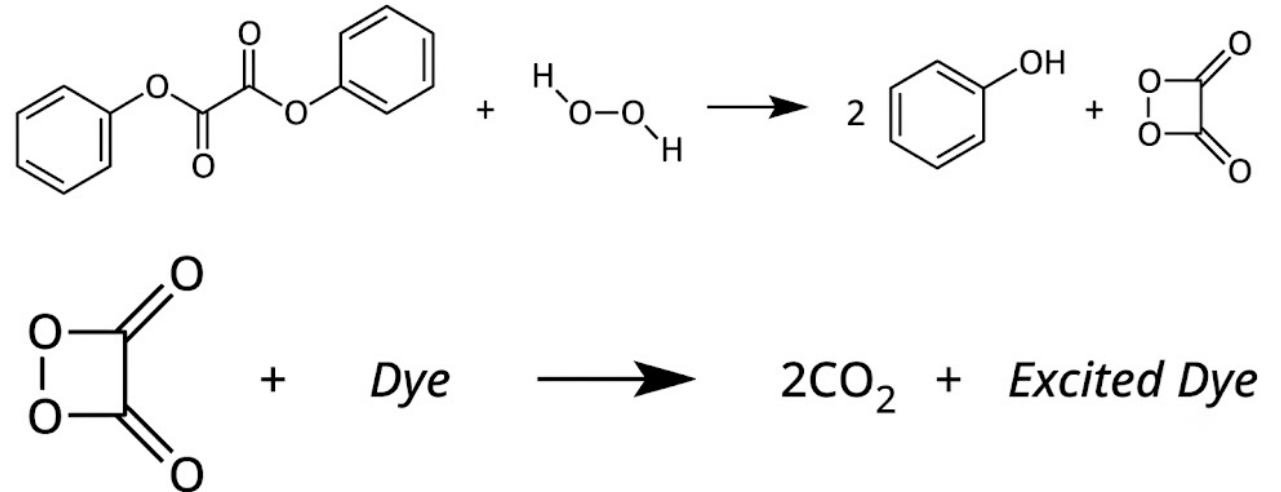
Example



Hot water

Cold water

Rate of the chemiluminescence reaction in light sticks (Cyalume light sticks).



1,2-dioxetanedione

Temperature and Rate

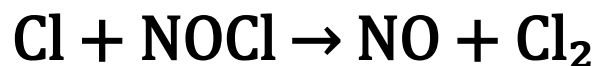
The Collision Model

- The **collision model** is based on the kinetic molecular theory.
 - ✓ **Molecules must collide to react:** more collisions per second = faster reaction rate.
 - ❑ **Higher concentration:** more particles → more collisions → faster reaction.
 - ❑ **Higher temperature:** particles move faster, collide more often and with greater energy → faster reaction.
- Not every collision causes a reaction: collisions must have the **right energy** and **orientation**.
- *In a chemical reaction, bonds are broken and new bonds are formed*

Temperature and Rate

The Orientation Factor

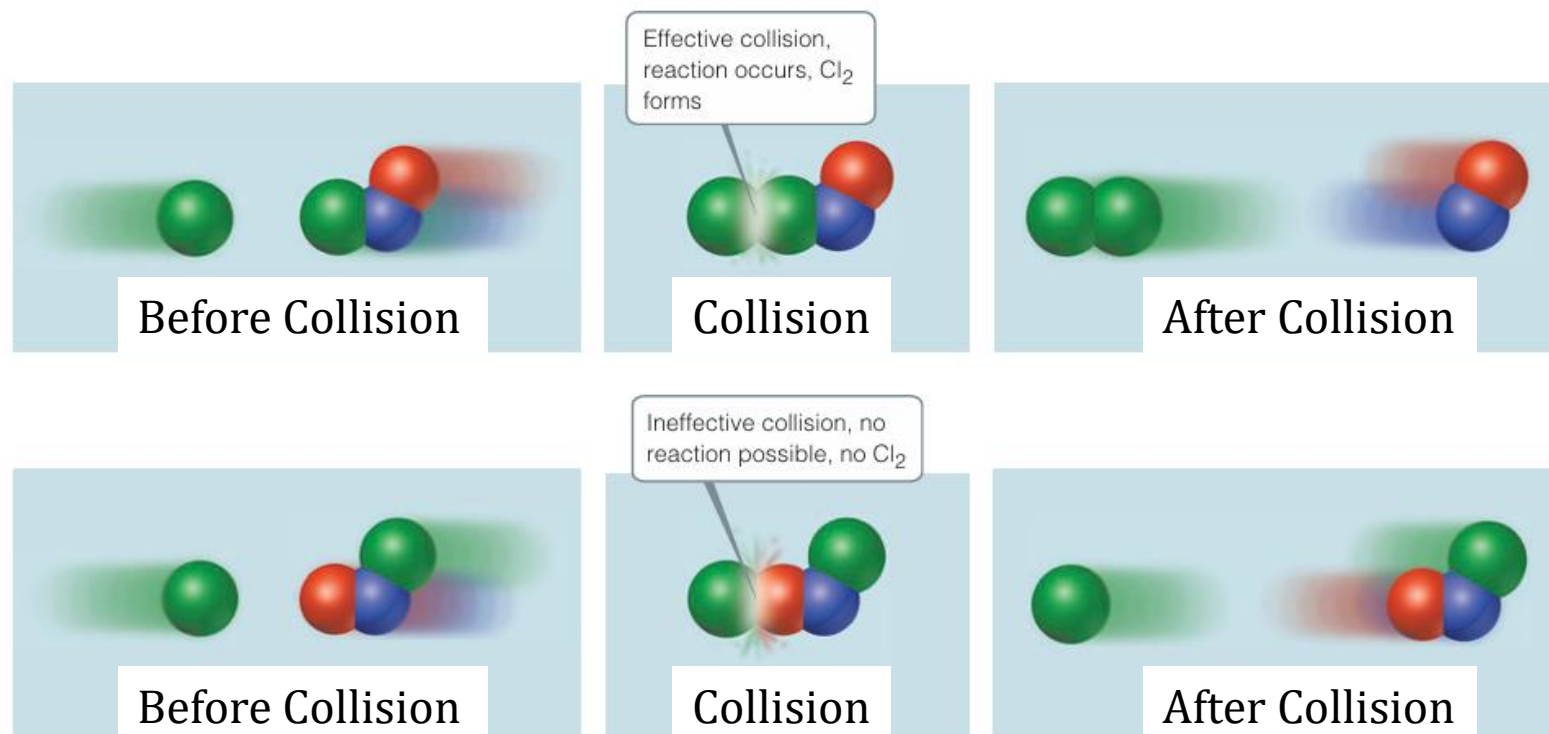
- Molecules can often collide without forming products.
- Aligning molecules properly can lead to chemical reactions.
- Bonds must be broken and made, and atoms need to be in proper positions.



- Cl: Chlorine atom
- NOCl: Nitrosyl chloride
- NO: Nitric oxide
- Cl₂: Chlorine gas

Q. Which collision process led to the reaction?

● : O ● : N ● : Cl

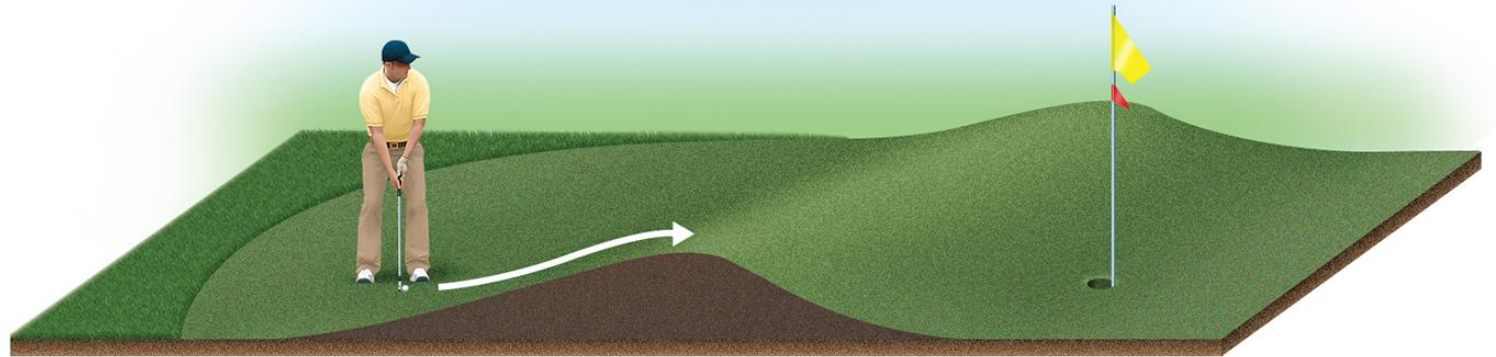


Temperature and Rate

Activation Energy (E_a)

- The minimum energy needed for a reaction to take place is called **activation energy** (Svante Arrhenius, 1888).
- An energy barrier must be overcome for a reaction to take place, much like the ball must be hit to overcome the barrier in the figure below.

Q. Where does the activation energy come from?

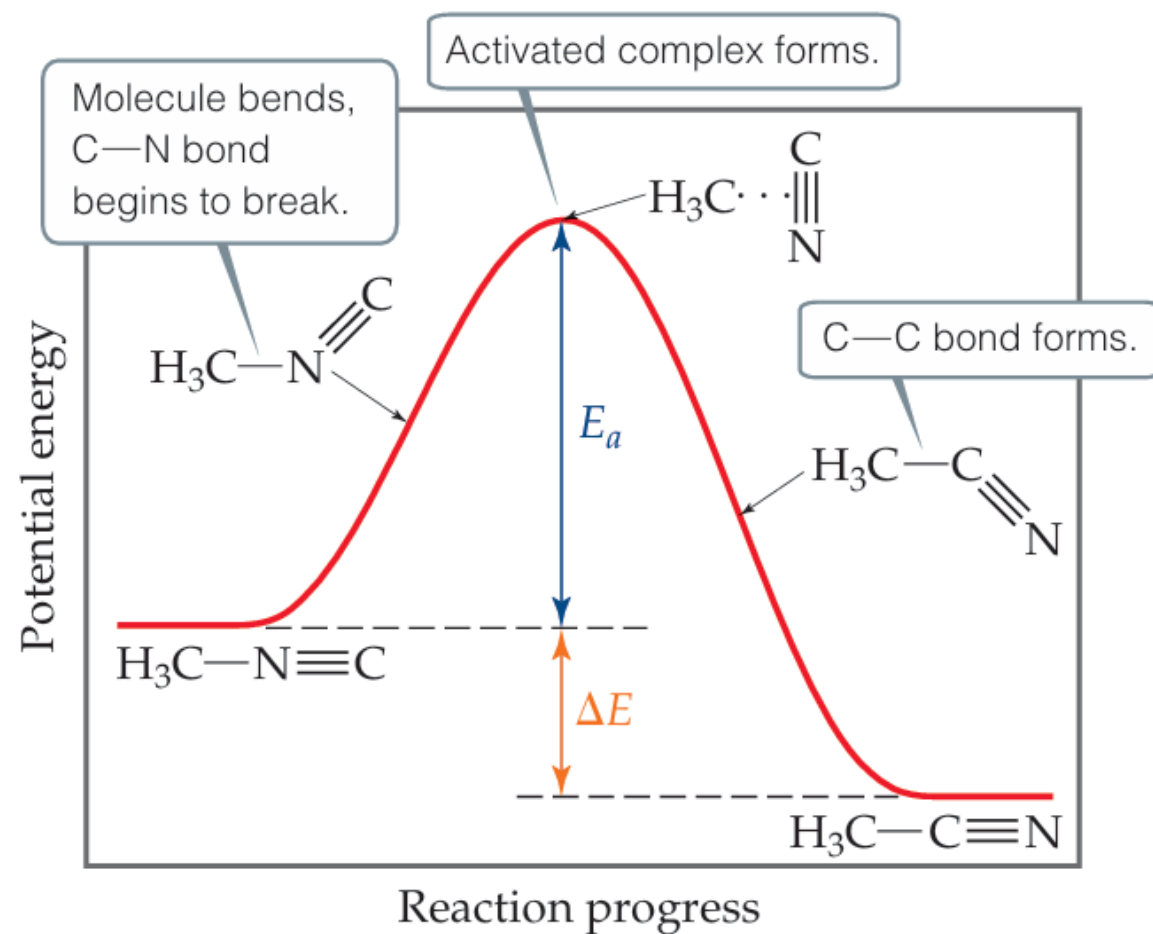
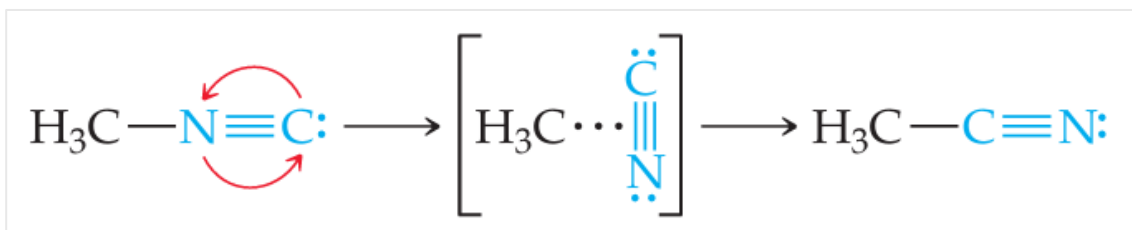


- *In a chemical reaction, bonds are broken and new bonds are formed*

Temperature and Rate

Activation Energy: transition state

- Reactants gain energy as the reaction proceeds until the particles reach the maximum energy state.
- The organization of the atoms at this highest energy state is called the **transition state** (or activated complex).
- The energy needed to form this state is called the activation energy.

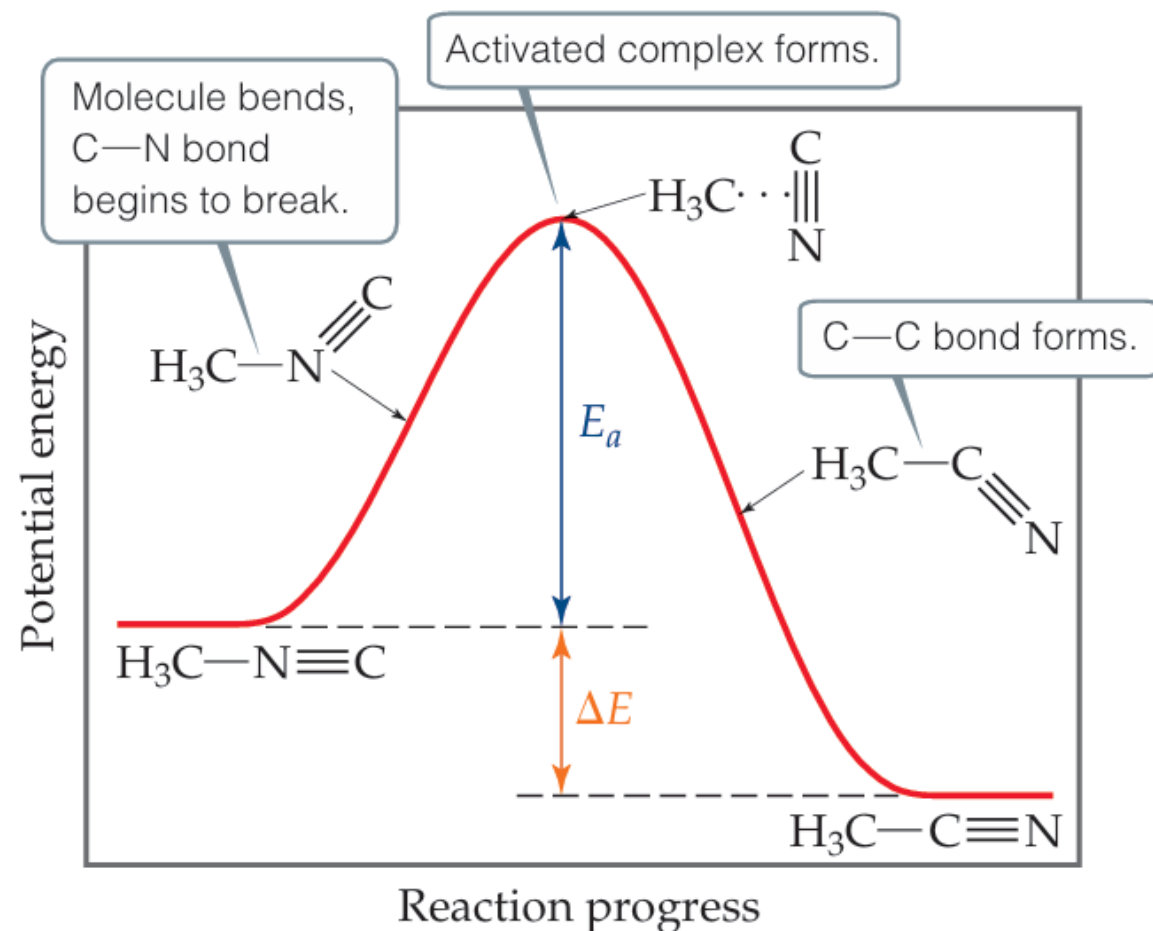


Energy profile for conversion of methyl isonitrile to its isomer acetonitrile.

Temperature and Rate

Activation Energy: transition state

- Reactions can be endothermic or exothermic after the **transition state**.
- Rate constant depends on the magnitude of E_a .
- The lower the value of E_a is, the larger the rate constant and the faster the reaction



Energy profile for conversion of methyl isonitrile to its isomer acetonitrile.

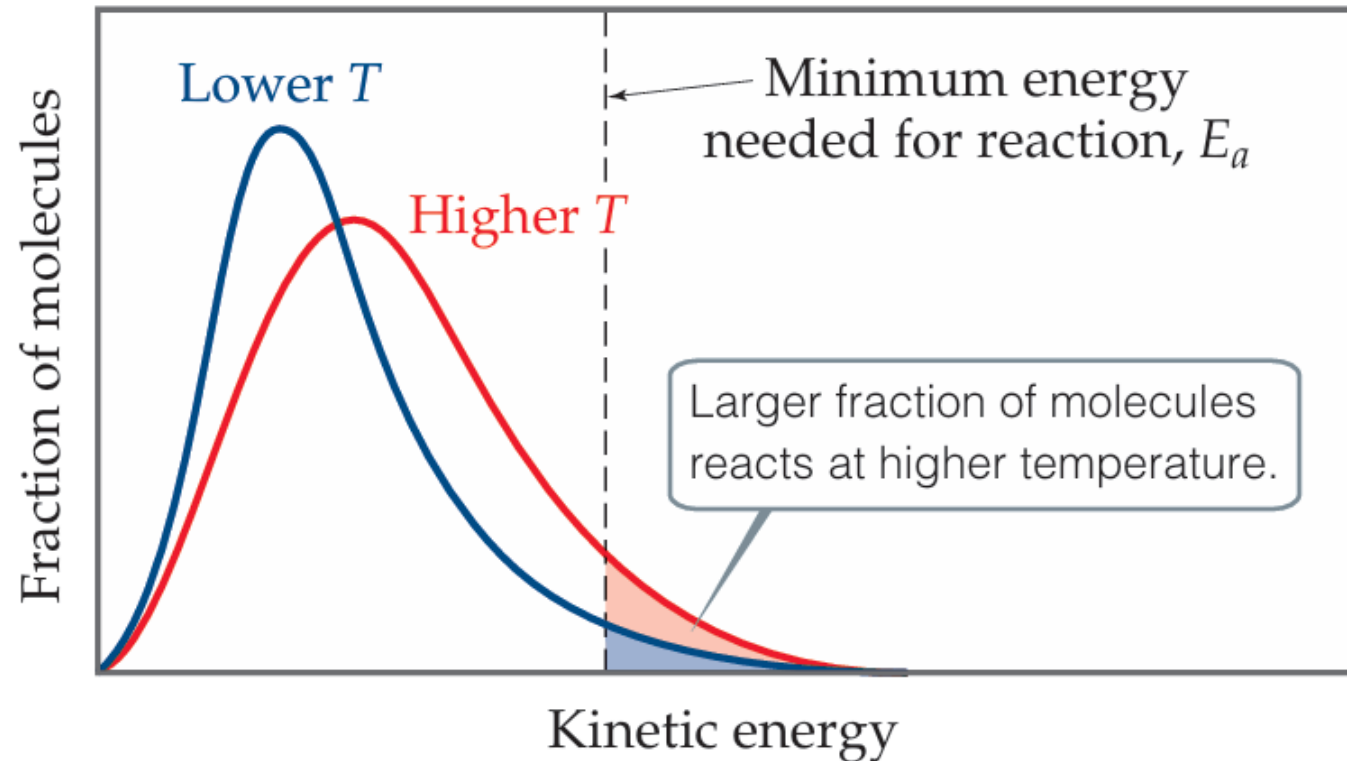
Temperature and Rate

Activation Energy: effect of temperature

- Gases have an average temperature, but each individual molecule has its own energy.
- At higher energies, more molecules possess the energy needed for the reaction to occur.

$$f = e^{-E_a/RT}$$

- ✓ R: gas constant
- ✓ T: absolute temperature
- ✓ f: fraction of gas molecules that have kinetic energy equal to or greater than E_a



The effect of temperature on the distribution of kinetic energies of molecules in a sample

Temperature and Rate

Activation Energy: The Arrhenius Equation

- For most reactions, the increase in rate with increasing temperature is nonlinear.
- Most reaction-rate data obeyed an equation based on:
 - a. the fraction of molecules possessing energy E_a or greater
 - b. the number of collisions per second, and
 - c. the fraction of collisions that have the appropriate orientation.

$$k = Ae^{-E_a/RT}$$

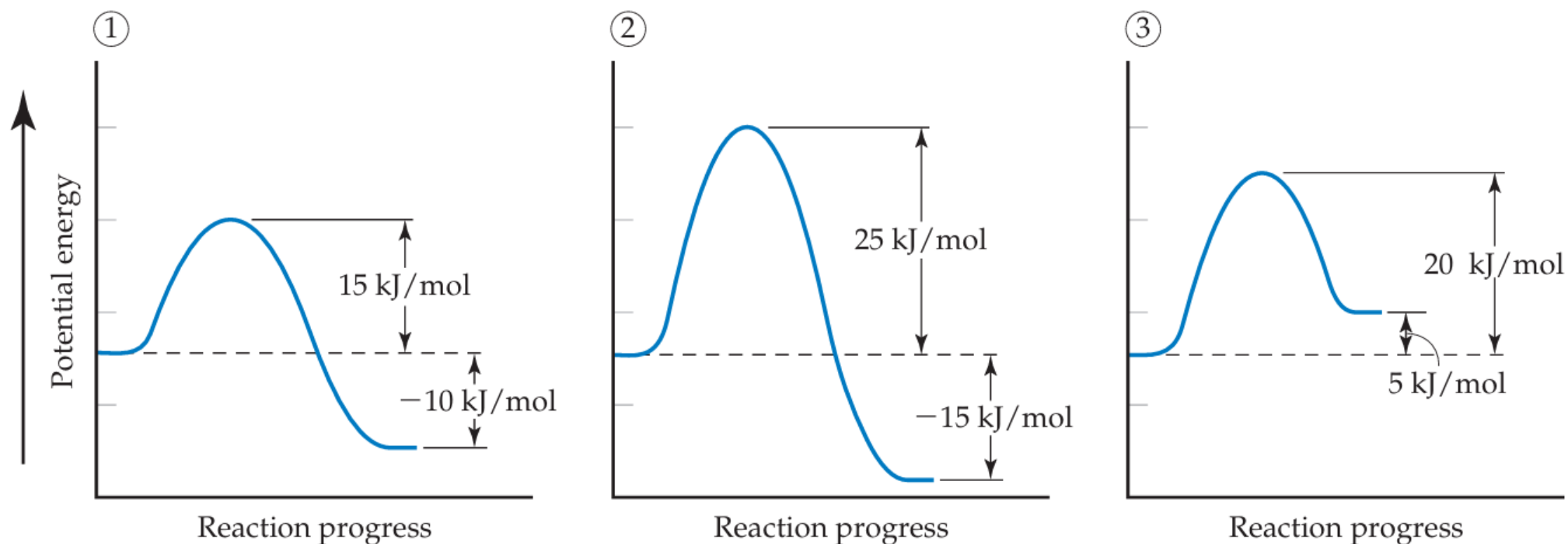
- ✓ k : rate constant
- ✓ E_a : activation energy
- ✓ R : gas constant (18.314 J/mol.K)
- ✓ T : absolute temperature
- ✓ A : frequency factor(constant)

At fixed values of T and A , the rate constant decreases as E_a increases.

Temperature and Rate

Activation Energy: The Arrhenius Equation

Q. Consider a series of reactions having these energy profiles. Rank the forward rate constants from smallest to largest assuming all three reactions have nearly the same value for the frequency factor A .



Temperature and Rate

Activation Energy: Determining the Activation Energy

$$k = Ae^{-E_a/RT}$$

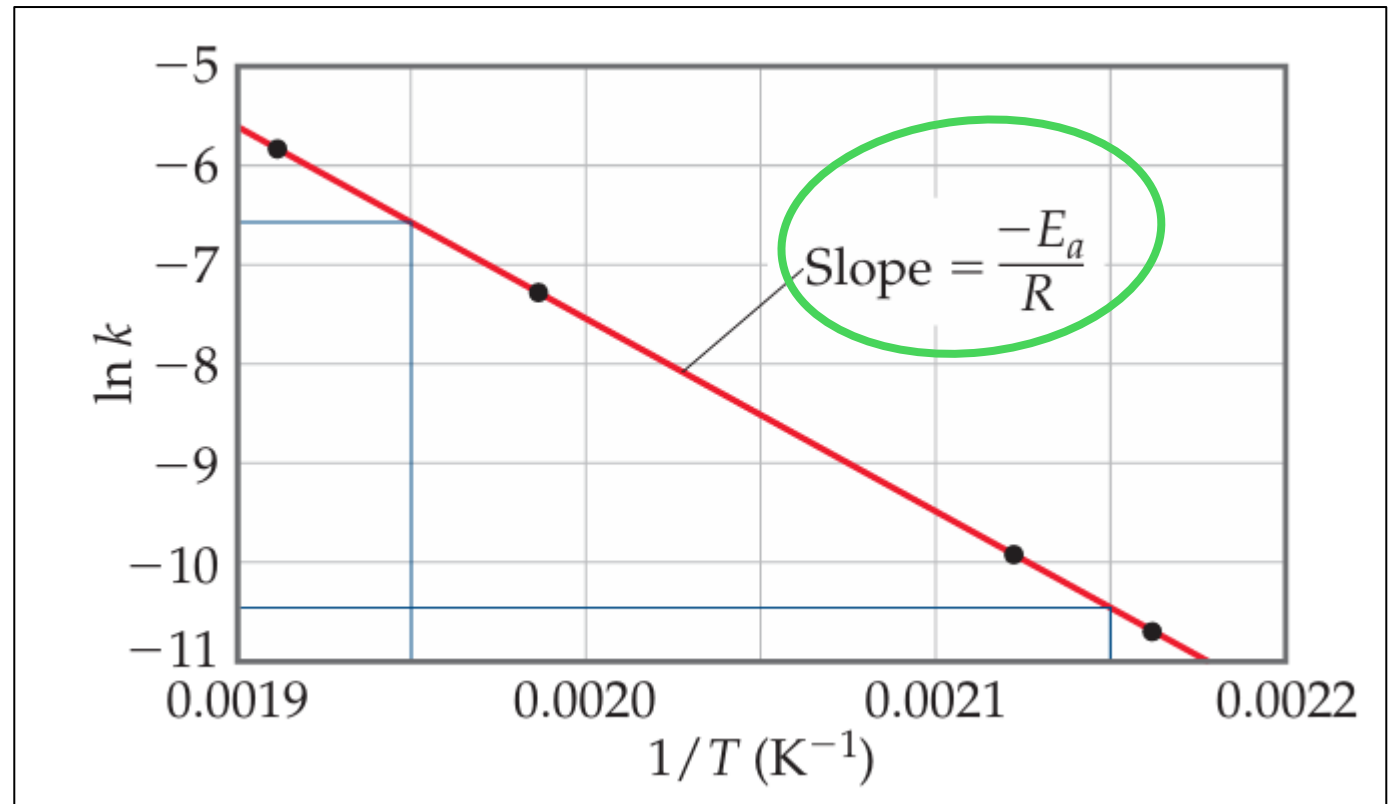
$$\ln k = \ln A e^{-E_a/RT}$$

$$\ln k = \ln e^{-E_a/RT} + \ln A$$

$$\ln k = -\frac{E_a}{RT} + \ln A$$



$$y = mx + b$$



Graphical determination of activation energy E_a

Temperature and Rate

Activation Energy: Determining the Activation Energy

$$k = Ae^{-E_a/RT}$$

- The activation energy can also be determined by measuring k at a series of temperatures.

$$\ln k = \ln A e^{-E_a/RT}$$

$$\ln k_1 = -\frac{E_a}{RT_1} + \ln A \quad \text{and} \quad \ln k_2 = -\frac{E_a}{RT_2} + \ln A$$

$$\ln k = \ln e^{-E_a/RT} + \ln A$$

$$\ln k = -\frac{E_a}{RT} + \ln A$$

$$\begin{array}{ccc} \updownarrow & \updownarrow & \updownarrow \\ y & = & mx + b \end{array}$$

$$\ln k_1 - \ln k_2 = \left(-\frac{E_a}{RT_1} + \ln A \right) - \left(-\frac{E_a}{RT_2} + \ln A \right)$$

$$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Temperature and Rate

Q. The rate constants for the rearrangement of methyl isonitrile at various temperatures are shown below (pp. 686).

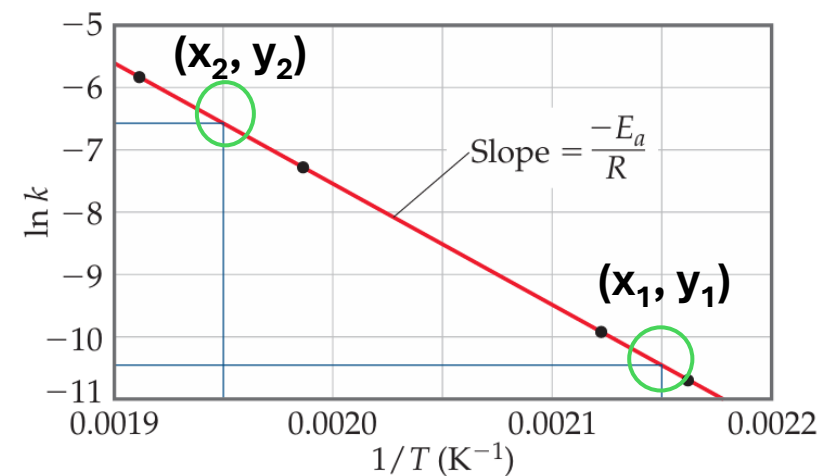
- Calculate the activation energy for the reaction.
- What is the value of the rate constant at 430.0 K?

T (°C)	k (s ⁻¹)	T (K)	1/T (K ⁻¹)	ln k
189.7	2.52×10^{-5}	462.9	2.160×10^{-3}	-10.589
198.9	5.25×10^{-5}	472.1	2.118×10^{-3}	-9.855
230.3	6.30×10^{-4}	503.5	1.986×10^{-3}	-7.370
251.2	3.16×10^{-3}	524.4	1.907×10^{-3}	-5.757

$$\text{Slope} = -\frac{E_a}{R}$$

$$\text{Slope} = \frac{\Delta y}{\Delta x}$$

$$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



$$(x_1, y_1) = (0.00215, -10.4)$$

$$(x_2, y_2) = (0.00195, -6.6)$$

Reaction Mechanisms

Q. How do we know the process of bond breaking and bond formation in a chemical reaction?

- The steps by which a reaction occurs is called the **reaction mechanism**.
- Reactions may occur all at once or through several discrete steps.

Elementary Reactions

- Reactions that occur in a single event or step
- Their molecularity can be unimolecular or bimolecular



Reaction Mechanisms

Elementary Reactions

TABLE 14.3 Elementary Reactions and Their Rate Laws

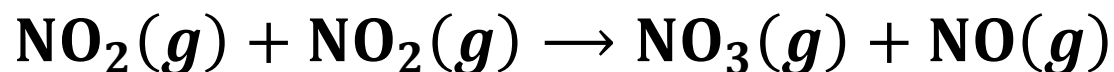
Molecularity	Elementary Reaction	Rate Law
Unimolecular	$A \longrightarrow \text{products}$	$\text{Rate} = k[A]$
Bimolecular	$A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^2$
Bimolecular	$A + B \longrightarrow \text{products}$	$\text{Rate} = k[A][B]$
Termolecular	$A + A + A \longrightarrow \text{products}$	$\text{Rate} = k[A]^3$
Termolecular	$A + A + B \longrightarrow \text{products}$	$\text{Rate} = k[A]^2[B]$
Termolecular	$A + B + C \longrightarrow \text{products}$	$\text{Rate} = k[A][B][C]$

- Nearly all reaction mechanisms contain only unimolecular and bimolecular elementary reactions.
- If a reaction is **elementary**, its rate law is based directly on its molecularity.

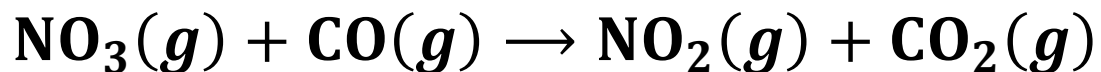
Reaction Mechanisms

Multistep Mechanisms

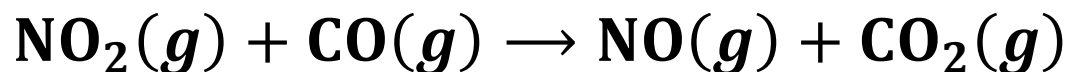
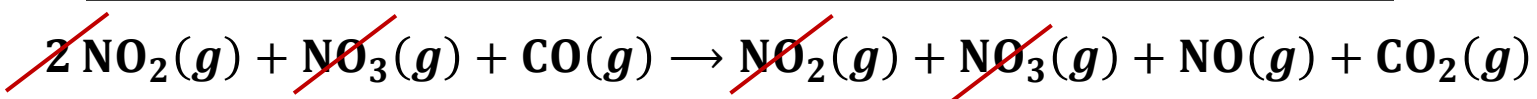
- Multistep mechanism consisting of a sequence of elementary reactions.
- The net change is represented by a balanced chemical equation.



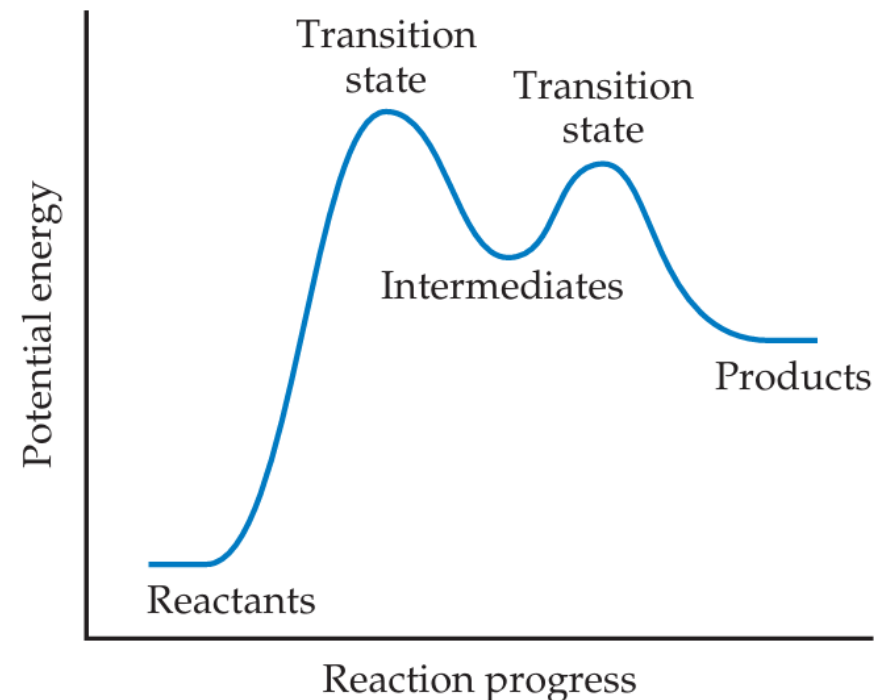
Nitrogen dioxide Nitrogen dioxide Nitrogen trioxide Nitric oxide



Nitrogen trioxide Carbon monoxide Nitrogen dioxide Carbon dioxide



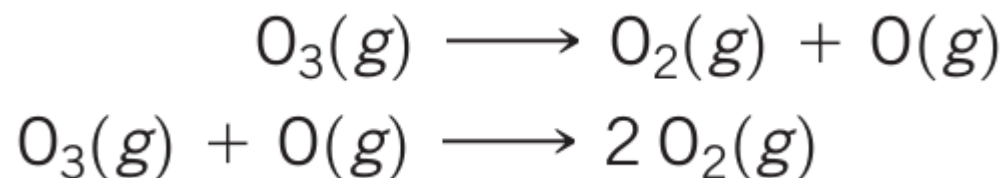
- NO_3 is an intermediate



The energy profile of a two-step reaction

Reaction Mechanisms

Q. It has been proposed that the conversion of ozone into O_2 proceeds by a two-step mechanism:

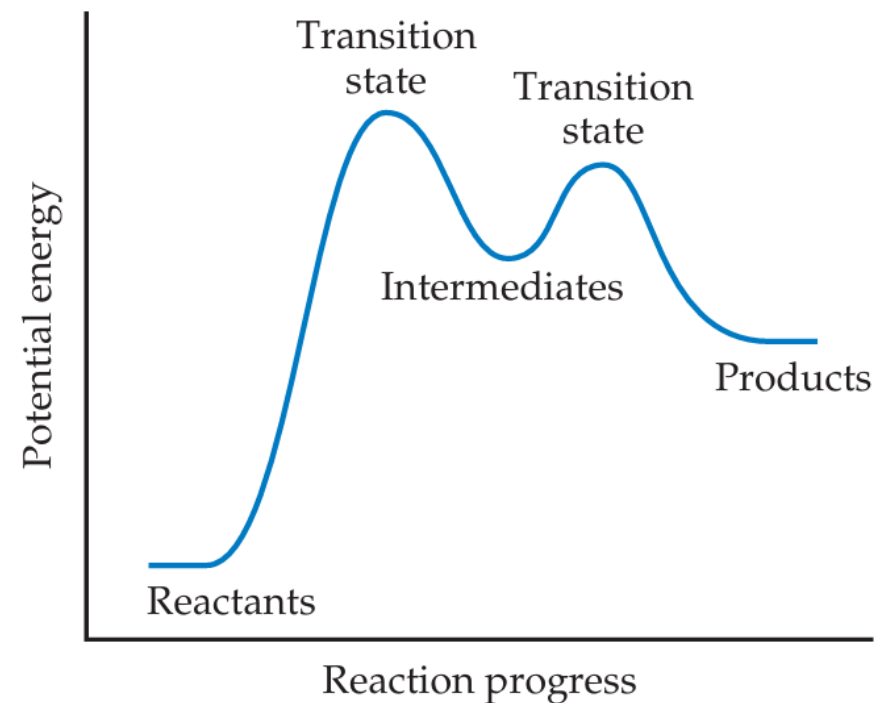


- (a) Describe the molecularity of each elementary reaction in this mechanism.
- (b) Write the equation for the overall reaction.
- (c) Identify the intermediate(s).

Reaction Mechanisms

Multistep Mechanisms: The Rate-Determining Step

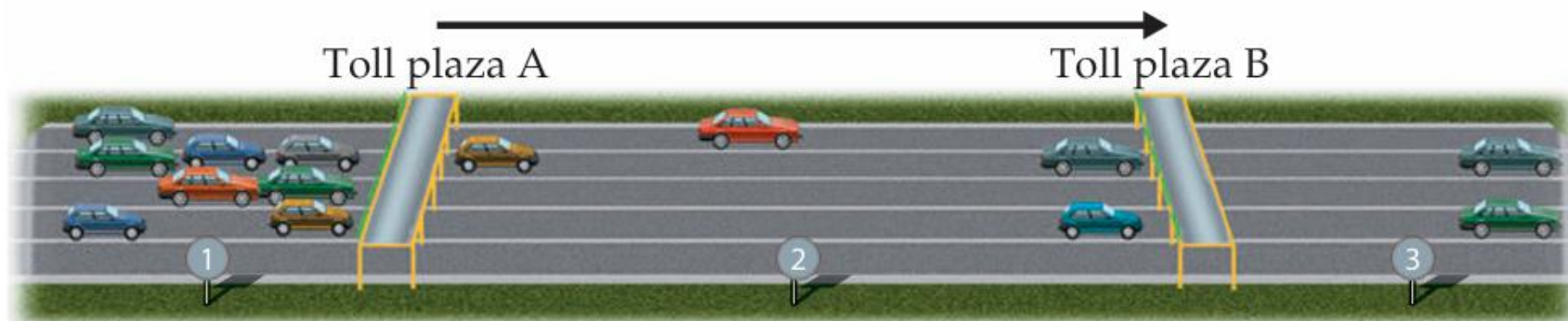
- Most reactions occur by mechanisms that involve two or more elementary reactions.
- Each step of the mechanism has its own rate constant and activation energy.
- Often one step is much slower than the others
- The overall rate of a reaction cannot exceed the rate of the slowest elementary step.
 - ✓ The slowest step in a multistep reaction determines the overall rate, and it is called the **rate-determining step** (or rate-limiting step).
 - ✓ The rate-determining step governs the rate law for the overall reaction



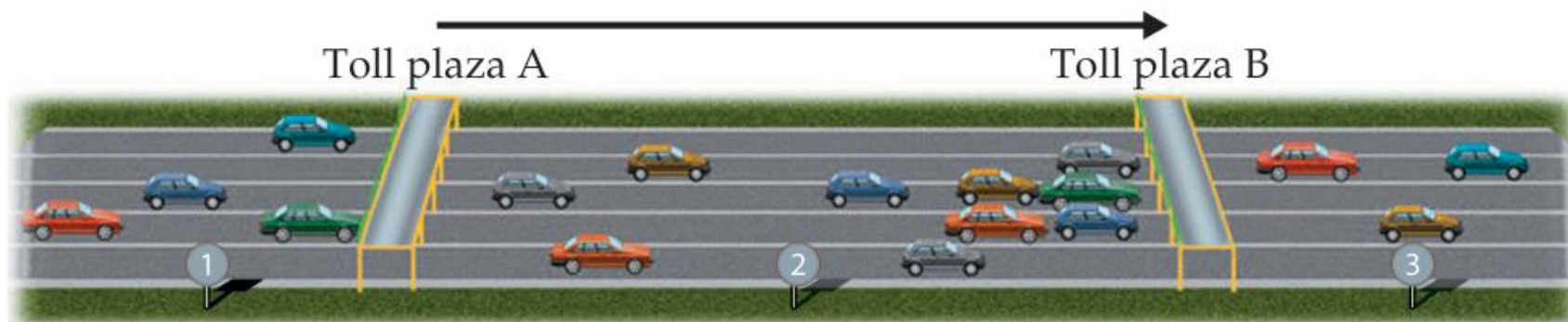
The energy profile of a two-step reaction

Reaction Mechanisms

Multistep Mechanisms: The Rate-Determining Step



(a) Cars slowed at toll plaza A, rate-determining step is passage through A



(b) Cars slowed at toll plaza B, rate-determining step is passage through B

Rate-determining steps in traffic flow on a toll road.

- Suppose that one or more gates at toll plaza A are malfunctioning

Step 1: Point 1 → Point 2 (through toll plaza A)

Step 2: Point 2 → Point 3 (through toll plaza B)

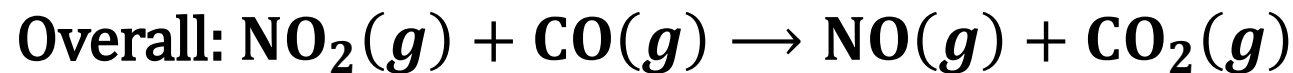
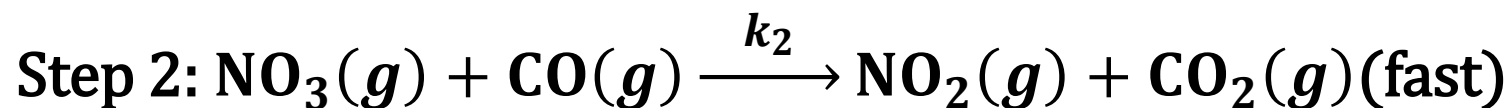
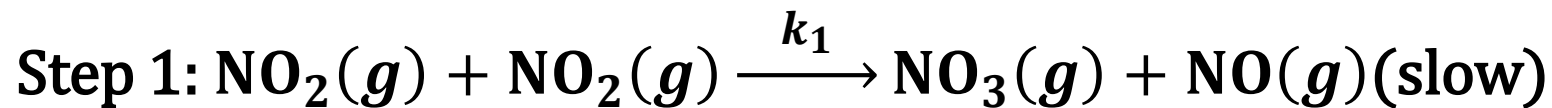
Overall: Point 1 → Point 3 (through both toll plazas)

- Suppose that one or more gates at toll plaza B are malfunctioning

Reaction Mechanisms

Multistep Mechanisms: Mechanisms with a Slow Initial Step

- Consider the reaction of NO_2 and CO to produce NO and CO_2
 - Below 225°C , it is found experimentally that the rate law for this reaction is second order in NO_2 and zero order in CO : **Rate = $k[\text{NO}_2]^2$**



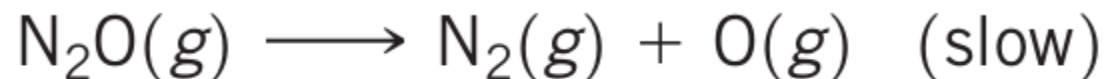
- Step 1 is slow and step 2 is fast ($k_2 \gg k_1$).
- Step 1 is the rate-determining step.

$$\text{Rate} = k_1[\text{NO}_2]^2$$

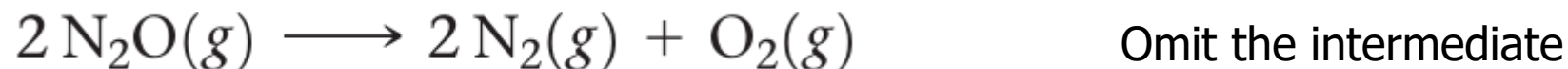
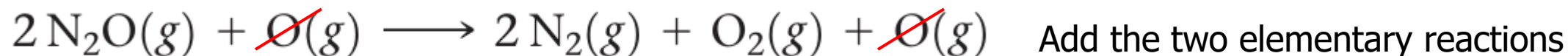
- Reaction intermediate does not appear in the rate equation.*

Reaction Mechanisms

- Q. The decomposition of nitrous oxide, N_2O , is believed to occur by a two-step mechanism:



- (a) Write the equation for the overall reaction.

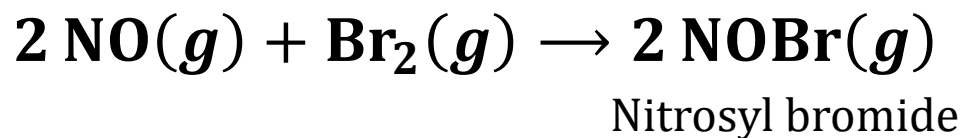


- (b) Write the rate law for the overall reaction.

Reaction Mechanisms

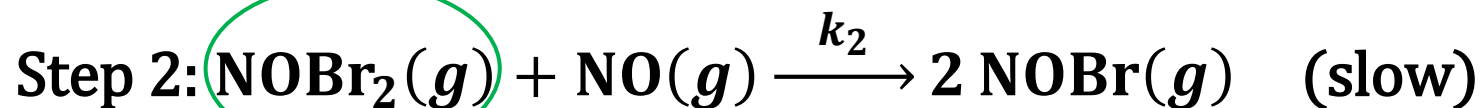
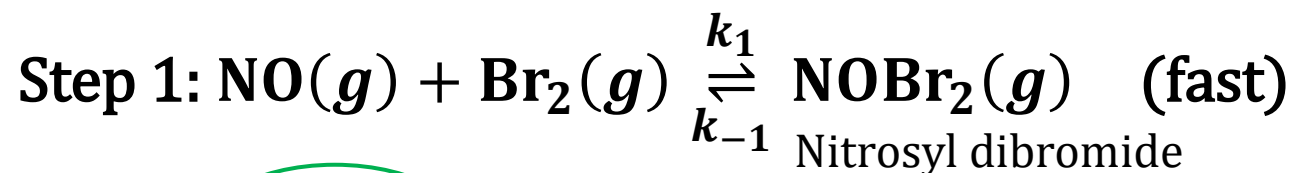
Multistep Mechanisms: Mechanisms with a Fast Initial Step

- Consider one example: the gas-phase reaction of nitric oxide with bromine



$$\text{Rate} = k[\text{NO}]^2[\text{Br}_2]$$

✓ **Experimentally
determined rate law**



$$\text{Rate} = k_2[\text{NOBr}_2][\text{NO}]$$

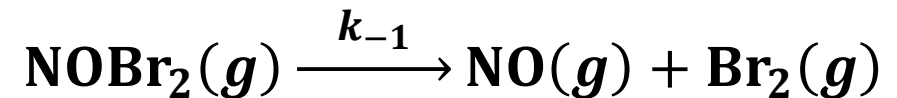
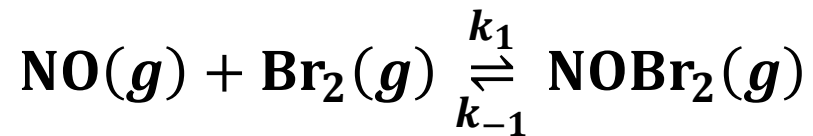
How to determine $[\text{NOBr}_2]$?

- With the aid of some assumptions, the concentration of the intermediate NOBr_2 can be expressed in terms of the concentrations of the starting reactants NO and Br_2 .

Reaction Mechanisms

Multistep Mechanisms: Mechanisms with a Fast Initial Step

- ✓ **Assumption 1:** NOBr_2 is slowly consumed in step 2 as soon as it is produced in step 1.
- ✓ **Assumption 2:** NOBr_2 decomposes in the reverse process of step 1 (equilibrium is established in the fast step).



$$\begin{array}{ccc} k_1[\text{NO}][\text{Br}_2] & = & k_{-1}[\text{NOBr}_2] \\ \text{Rate of forward reaction} & & \text{Rate of reverse reaction} \end{array}$$

$$[\text{NOBr}_2] = \frac{k_1}{k_{-1}} [\text{NO}][\text{Br}_2]$$

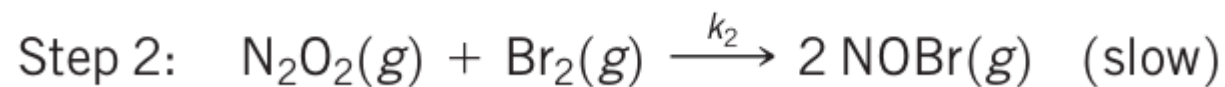
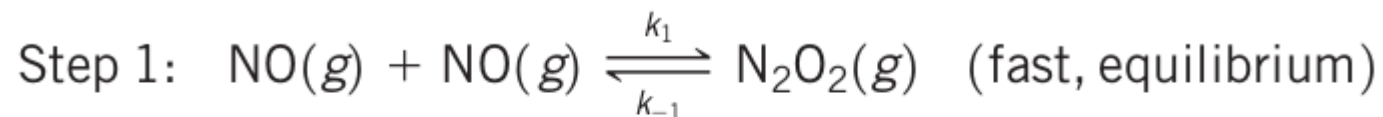
$$\text{Rate} = k_2[\text{NOBr}_2][\text{NO}]$$

$$\text{Rate} = k_2 \frac{k_1}{k_{-1}} [\text{NO}][\text{Br}_2][\text{NO}] = k[\text{NO}]^2[\text{Br}_2]$$

- ✓ Where the experimental rate constant k equals k_2k_1/k_{-1} .

Reaction Mechanisms

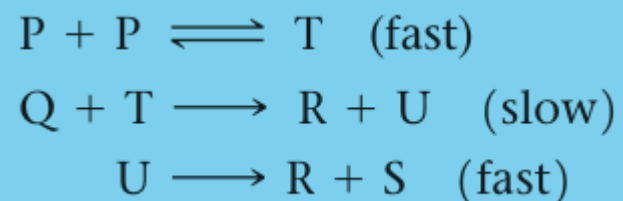
Q. Consider the following mechanism:



Show the steps to drive the following rate equation

$$\text{Rate} = k_2 \frac{k_1}{k_{-1}} [\text{NO}]^2 [\text{Br}_2] = k [\text{NO}]^2 [\text{Br}_2]$$

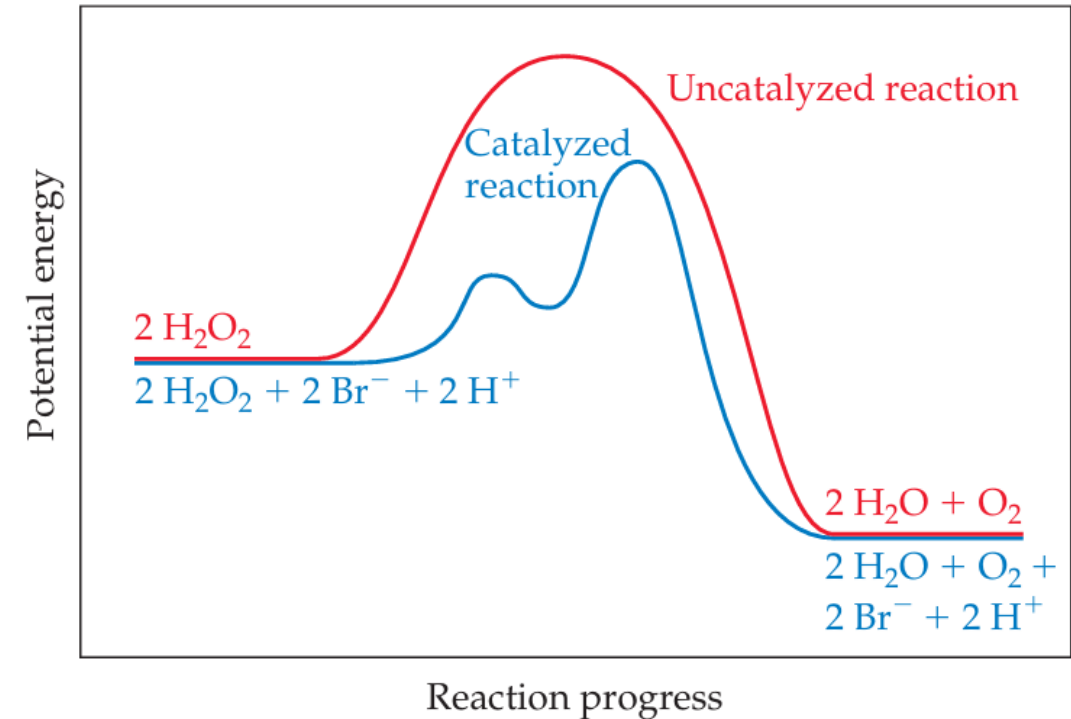
Q. Consider the following hypothetical reaction: $2 \text{P} + \text{Q} \rightarrow 2 \text{R} + \text{S}$. Predict the rate law for this reaction if the reaction has the following mechanism.



Catalysis

Q. How do catalysts affect the rate of a reaction?

- Catalysts change the speed of a chemical reaction without undergoing a permanent chemical change themselves.
- A catalyst provides a different mechanism for the chemical reaction of lower overall activation energy.
- There are three types of catalysts:
 - 1) Homogeneous catalysts
 - 2) Heterogeneous catalysts
 - 3) Enzymes (Biological catalysts)

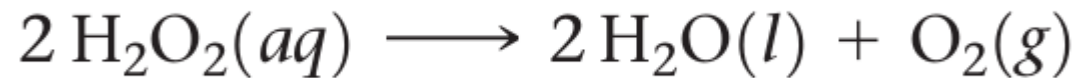


Energy profiles for the uncatalyzed and bromide-catalyzed decomposition of H_2O_2

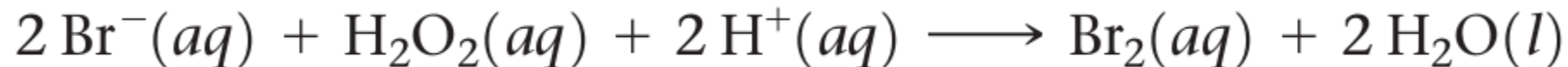
Catalysis-Types of catalysts

Homogeneous Catalysis

- **Homogeneous catalyst:** A catalyst that is present in the same phase as the reactants.
- Example: Uncatalyzed decomposition of hydrogen peroxide (slow):

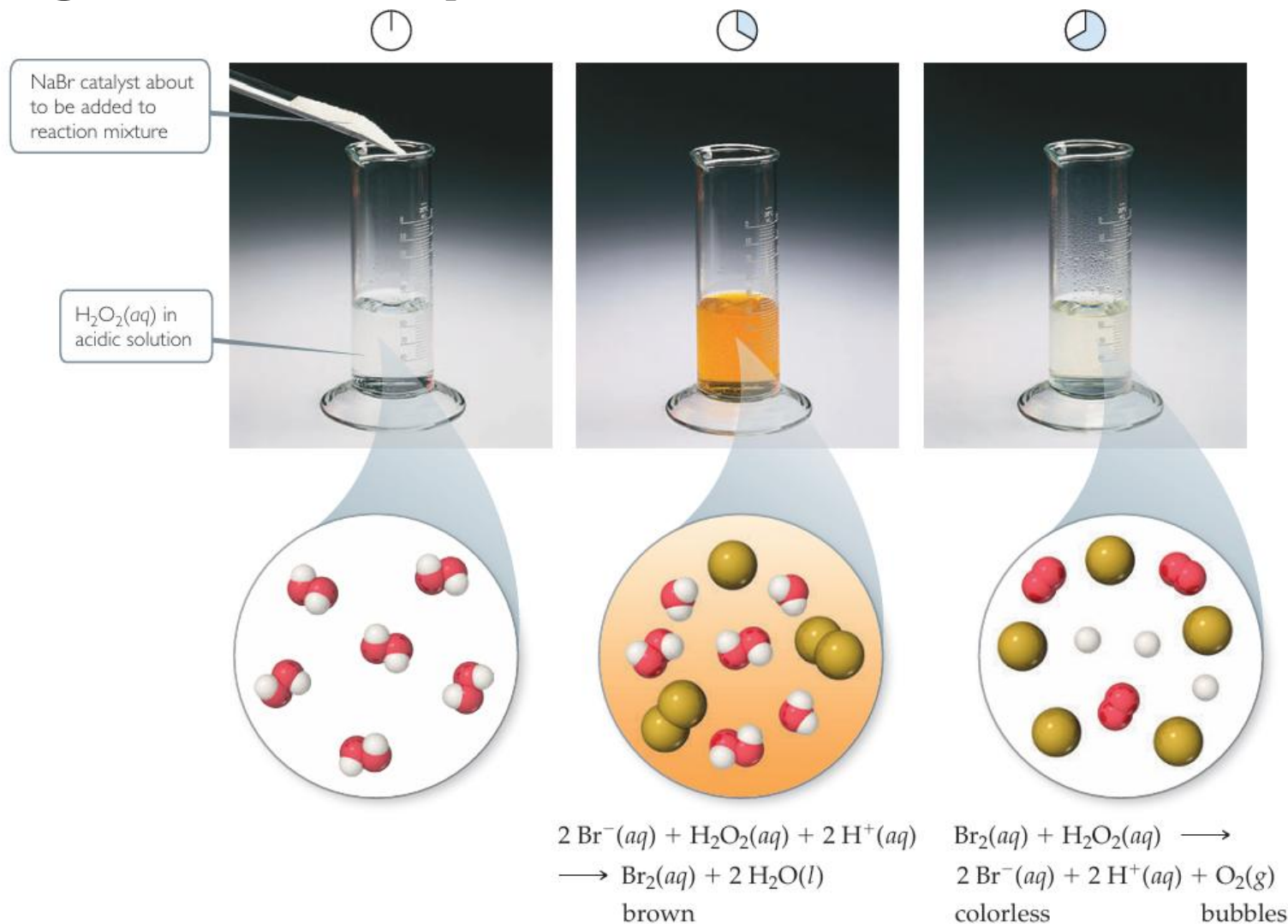


- Catalyzed decomposition of hydrogen peroxide (using bromide ion):



Catalysis-Types of catalysts

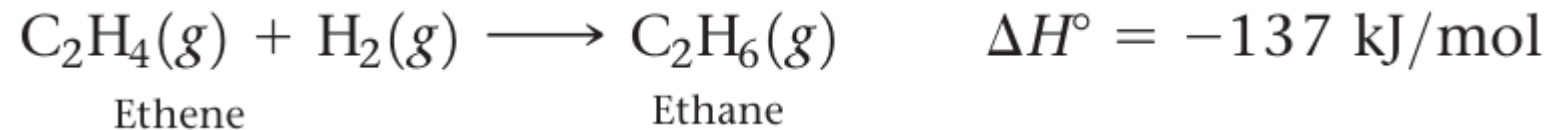
Homogeneous Catalysis



Catalysis-Types of catalysts

Heterogeneous Catalysis

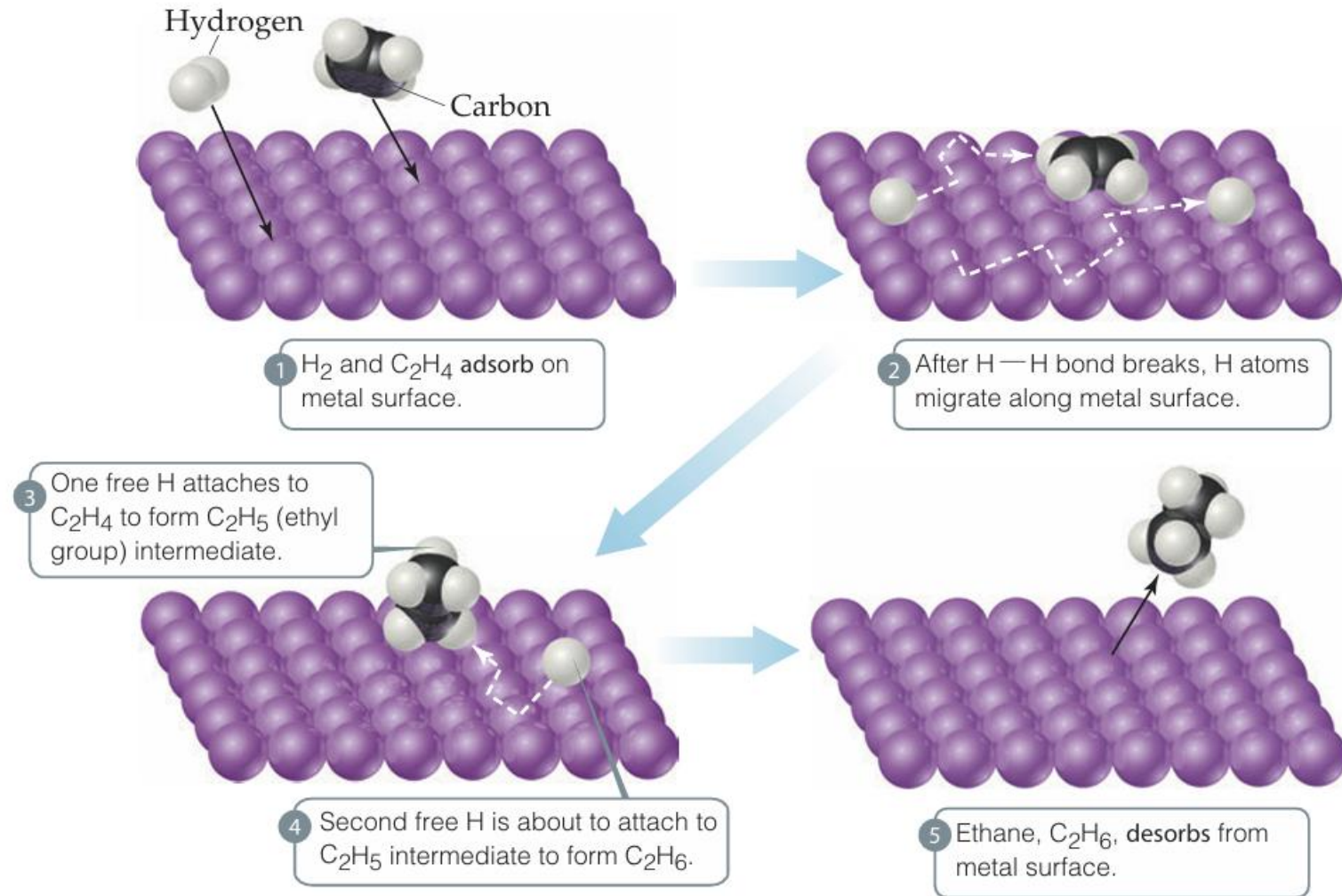
- **Heterogeneous catalyst:** exists in a phase different from the phase of the reactant molecules (usually as a solid in contact with either gaseous reactants or reactants in a liquid solution).
- Example: Uncatalyzed reaction of hydrogen gas with ethene gas (slow, exothermic):



- In the presence of a finely powdered metal (such as nickel, palladium, or platinum), the reaction occurs easily at room temperature.
- The initial step in heterogeneous catalysis is usually **adsorption** of reactants (the binding of molecules to the catalyst surface).

Catalysis-Types of catalysts

Heterogeneous Catalysis

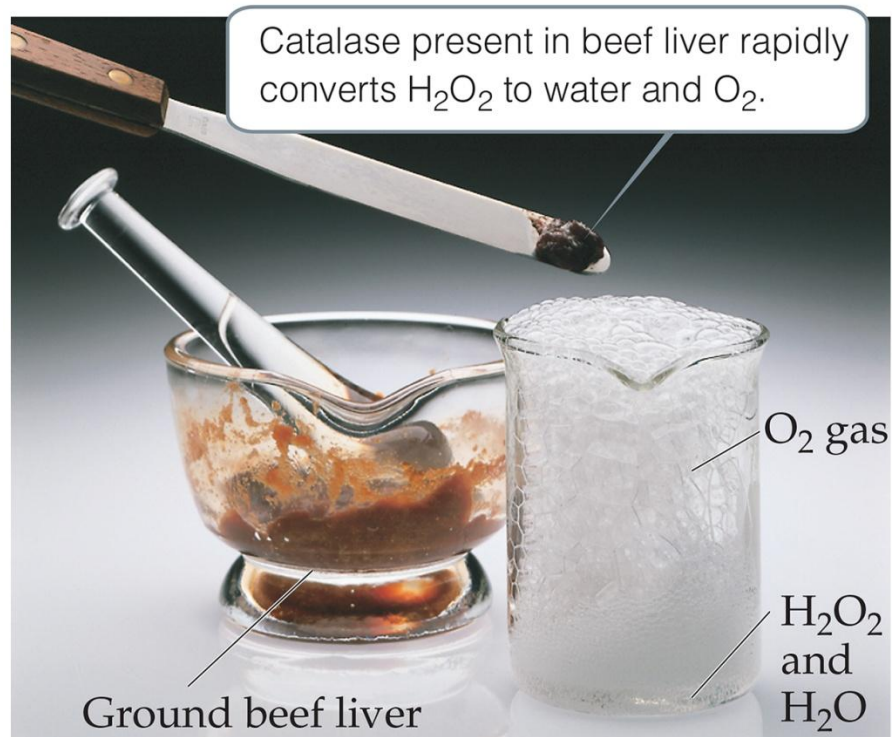


Heterogeneous catalysis.
Mechanism for the reaction
of ethene with hydrogen on
a catalytic surface.

Catalysis-Types of catalysts

Enzymes

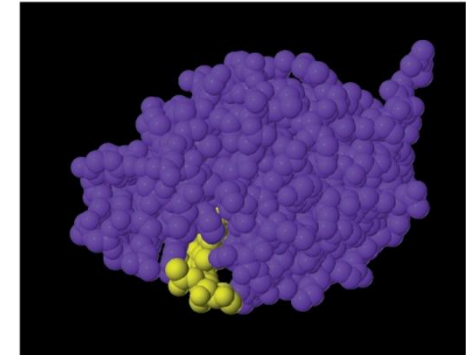
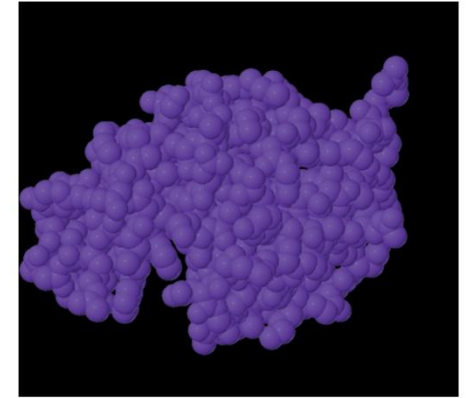
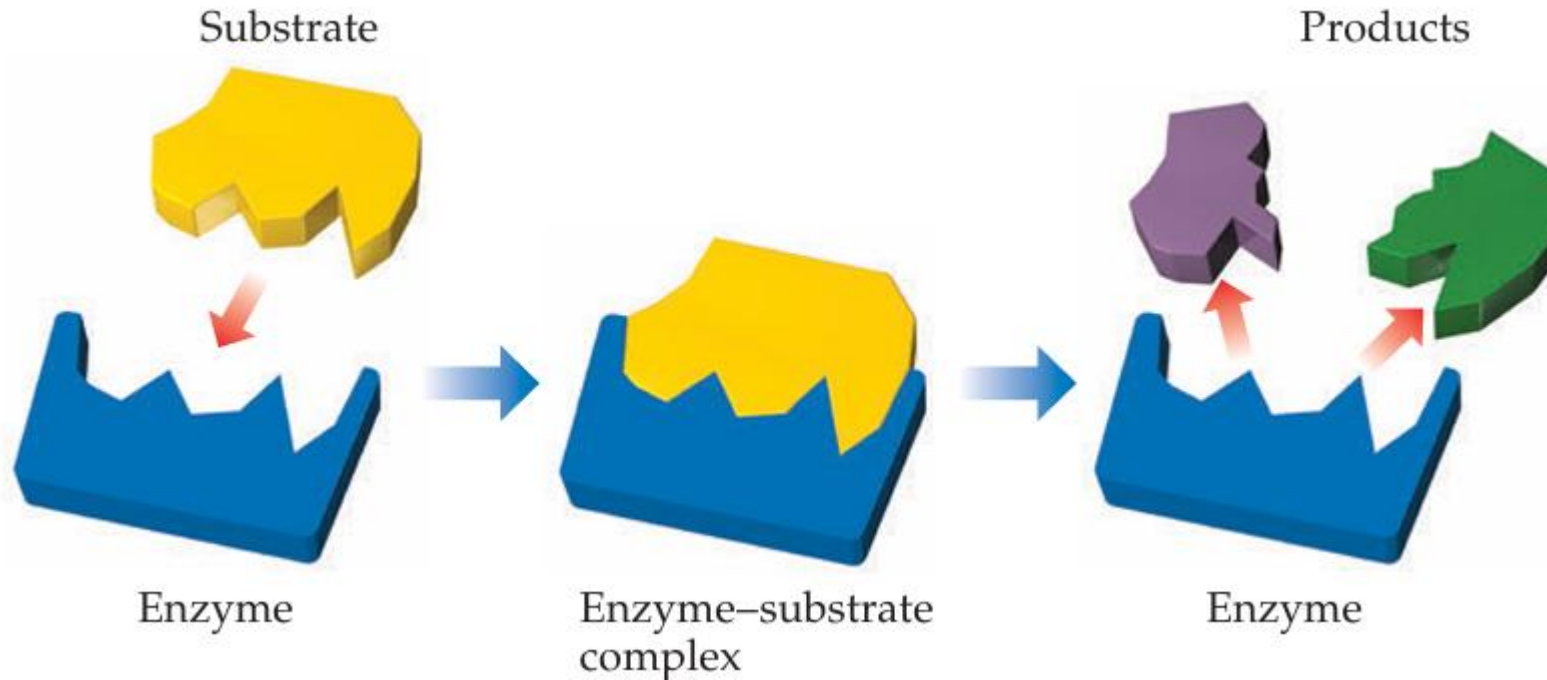
- **Enzymes** are biological catalysts.
- They have a region where the reactants attach. That region is called the **active site**.
- The reactants are referred to as **substrates**.



Enzymes speed up reactions.

Catalysis-Types of catalysts

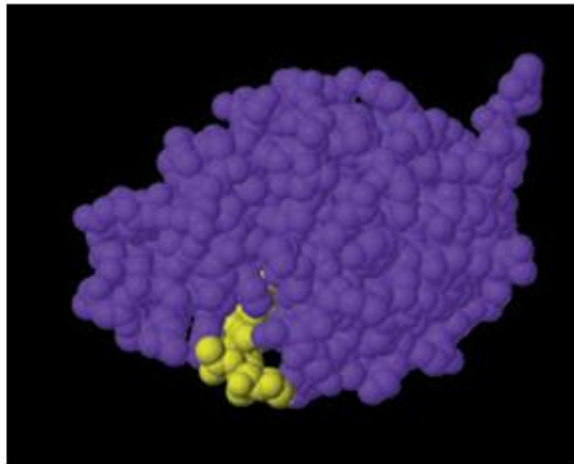
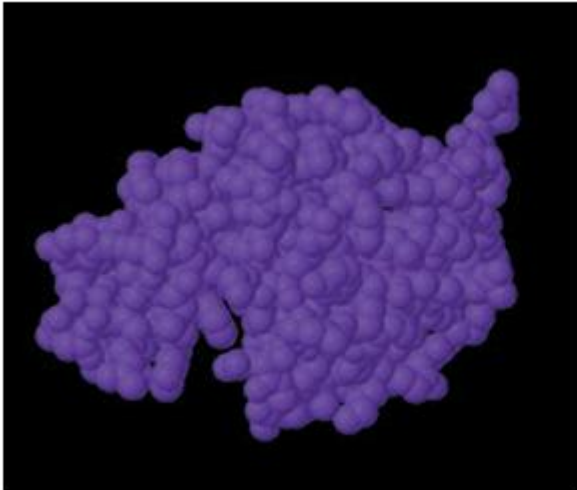
Enzymes: lock-and-key model



- In the enzyme-substrate model, the substrate fits into the active site of an enzyme, much like a key fits into a lock.
- They are specific.

Catalysis-Types of catalysts

Enzymes: lock-and-key model



- ✓ Lysozyme is important to the functioning of our immune system.
- ✓ Lysozyme accelerates reactions that damage (or “lyse”) bacterial cell walls.

A model of the enzyme lysozyme without and with a bound substrate molecule. Lysozyme was one of the first enzymes for which a structure–function relationship was described. This model shows how the substrate (yellow) “fits” into the active site of the enzyme

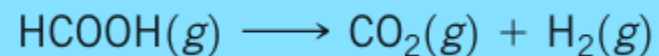
Catalysis-Types of catalysts

Enzyme Inhibition & Catalytic Efficiency

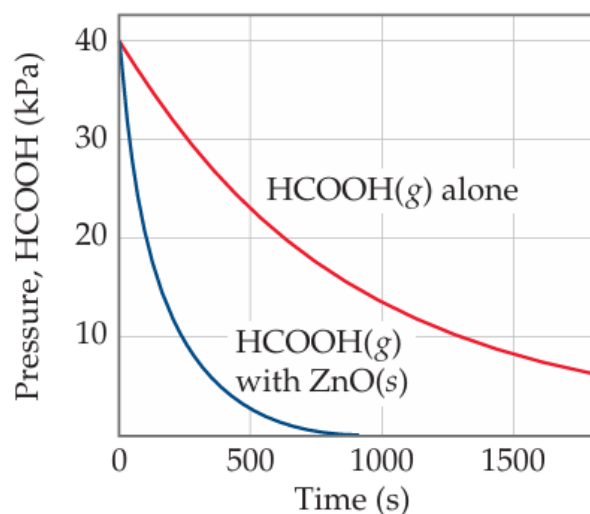
- **Enzyme inhibitors:** Molecules that reduce enzyme activity by interfering with the enzyme's active site.
- Two inhibition mechanisms:
 - ✓ **Active-site blocking:** Inhibitor prevents substrate from binding.
 - ✓ **Allosteric distortion:** Inhibitor binds elsewhere and changes the shape of the active site.
- Examples: Nerve poisons and toxic metals such as lead (Pb) and mercury (Hg).
- **High catalytic efficiency:** Enzymes can catalyze 10^3 – 10^7 reactions/second at one active site (turnover number).
- **Major advantage:** Enzymes can increase reaction rates by ≥ 1 million-fold compared with simple chemical catalysts due to their very low activation energies.

Summary Q

Q. Formic acid (HCOOH) decomposes in the gas phase at elevated temperatures as follows:



The uncatalyzed decomposition reaction is determined to be first order. A graph of the partial pressure of HCOOH versus time for decomposition at 838 K is shown as the red curve in the following Figure. When a small amount of solid ZnO is added to the reaction chamber, the partial pressure of acid versus time varies as shown by the blue curve in the Figure.



- Estimate the half-life and first-order rate constant for formic acid decomposition.
- What can you conclude from the effect of added ZnO on the decomposition of formic acid?
- The progress of the reaction was followed by measuring the partial pressure of formic acid vapor at selected times. Suppose that, instead, we had plotted the concentration of formic acid in units of mol/L . What effect would this have had on the calculated value of k ?
- The pressure of formic acid vapor at the start of the reaction is 40 kPa. Assuming constant temperature and ideal-gas behavior, what is the pressure in the system at the end of the reaction? If the volume of the reaction chamber is 436 cm^3 , how many moles of gas occupy the reaction chamber at the end of the reaction?
- The standard heat of formation of formic acid vapor is $\Delta H_f^\circ = -378.6 \text{ kJ/mol}$. Calculate ΔH° for the overall reaction. If the activation energy (E_a) for the reaction is 184 kJ/mol , sketch an approximate energy profile for the reaction, and label E_a , ΔH° , and the transition state.

PP. 702-703

By now, you should be able to:

- ✓ List the factors that affect the rate of chemical reactions.
- ✓ Determine the rate of a reaction by measuring the concentration of a reactant or product as a function of time.
- ✓ Relate the rate of formation of products and the rate of disappearance of reactants given the balanced chemical equation for the reaction.
- ✓ Explain the form and meaning of a rate law, including the ideas of reaction order and rate constant.
- ✓ Determine the rate law and rate constant for a reaction from a series of experiments given the measured rates for various concentrations of reactants.
- ✓ Apply the integrated form of a rate law to determine the concentration of a reactant at a given time.
- ✓ Apply the relationship between the rate constant of a first-order reaction and its half-life.
- ✓ Determine the order of a reaction from a table of reactant concentration as a function of time.
- ✓ Explain how the activation energy affects a rate and be able to use the Arrhenius equation.
- ✓ Predict a rate law for a reaction having a multistep mechanism given the individual steps in the mechanism.
- ✓ Explain the principles underlying catalysis.

Summary: Key equations:

- $$\text{Rate} = -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t} \quad [14.4]$$

Definition of reaction rate in terms of the components of the balanced chemical equation $a A + b B \longrightarrow c C + d D$
- $$\text{Rate} = k[A]^m[B]^n \quad [14.7]$$

General form of a rate law for the reaction $A + B \longrightarrow \text{products}$
- $$\ln[A]_t - \ln[A]_0 = -kt \quad \text{or} \quad \ln \frac{[A]_t}{[A]_0} = -kt \quad [14.12]$$

The integrated form of a first-order rate law for the reaction $A \longrightarrow \text{products}$
- $$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0} \quad [14.14]$$

The integrated form of a second-order rate law for the reaction $A \longrightarrow \text{products}$
- $$[A]_t = -kt + [A]_0 \quad [14.16]$$

The integrated form of a zero-order rate law for the reaction $A \longrightarrow \text{products}$
- $$t_{1/2} = \frac{0.693}{k} \quad [14.17]$$

Relating the half-life and rate constant for a first-order reaction
- $$k = Ae^{-E_a/RT} \quad [14.21]$$

The Arrhenius equation, which expresses how the rate constant depends on temperature
- $$\ln k = -\frac{E_a}{RT} + \ln A \quad [14.22]$$

Logarithmic form of the Arrhenius equation